Supplementary Information: Annex D – Structural Equation Modelling

Jardim et al. (2026). Trait-based insights into complexity–diversity relationships: different habitat complexity components drive niche diversity and community size. Hydrobiologia.

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Details on structural equation modelling

To determine and quantify the direct and indirect pathways linking habitat complexity to diversity across changing environments, a pSEM approach was used (Lefcheck 2016) . A set of causal path models was constructed from the hypothetical relationships identified in the literature among hydrosedimentary dynamics, habitat complexity facets of maerl and polychaete functional and taxonomic diversity (Figure 1 — main text, detailed in section 'Theoretical causal pathways').
Each observation (a sampling point at a given year, n = 90) was characterised in the pSEM in terms of 1) its position along the first axis of the hydrosedimentary stress PCA; 2) its median value for each habitat complexity metric — with nodule surface and nodule biomass as proxies of habitat amount, nodule convexity as a proxy of habitat geometry and nodule trait dispersion as a proxy of habitat diversity; and 3) total density and species richness of its polychaete assemblage as proxies of community size.
Details on the fit of each causal path model and final pSEM selection are described below.

Path models

All path models were visually checked for linear model assumptions, and tested for homoscedasticity and normality of residuals.
Total density of polychaetes, nodule surface and nodule trait dispersion were log transformed, and nodule biomass was square root transformed to linearise relationships and meet model assumptions. Convexity was also log-transformed; however, homoscedasticity and normality of residuals were not fully achieved for this variable within the piecewise structural equation model (pSEM). To assess the robustness of our results to this assumption violation, we fitted an alternative pSEM in which convexity was modelled as a response variable using a generalized linear model with a Gamma distribution and inverse link, applied to untransformed convexity values. Overall pSEM comparison based on Chi-squared statistics indicated similar global fit between the Gaussian and Gamma-based pSEMs. Focusing specifically on the convexity sub-model, predicted values from both approaches did not differ significantly (mean difference not significantly different from zero) and showed comparable predictive accuracy (RMSE and MAE; (Chai and Draxler 2014)).
Because both approaches yielded equivalent predictions and model fit, we retained the Gaussian pSEM for the main analyses in order to facilitate comparison of standardized path coefficients across model components. Importantly, qualitative conclusions regarding the direction and relative strength of paths were consistent across both model formulations.
Following are the diagnostic plots and statistical tests for each path model, grouped by response variable.

Habitat amount

Biomass

Diagnostic plots and tests for the path model $Biomass \sim Hydrosedimentary\ stress$ on the raw and square-root transformed response variable. Normality of residuals is only met after square-root transformation.

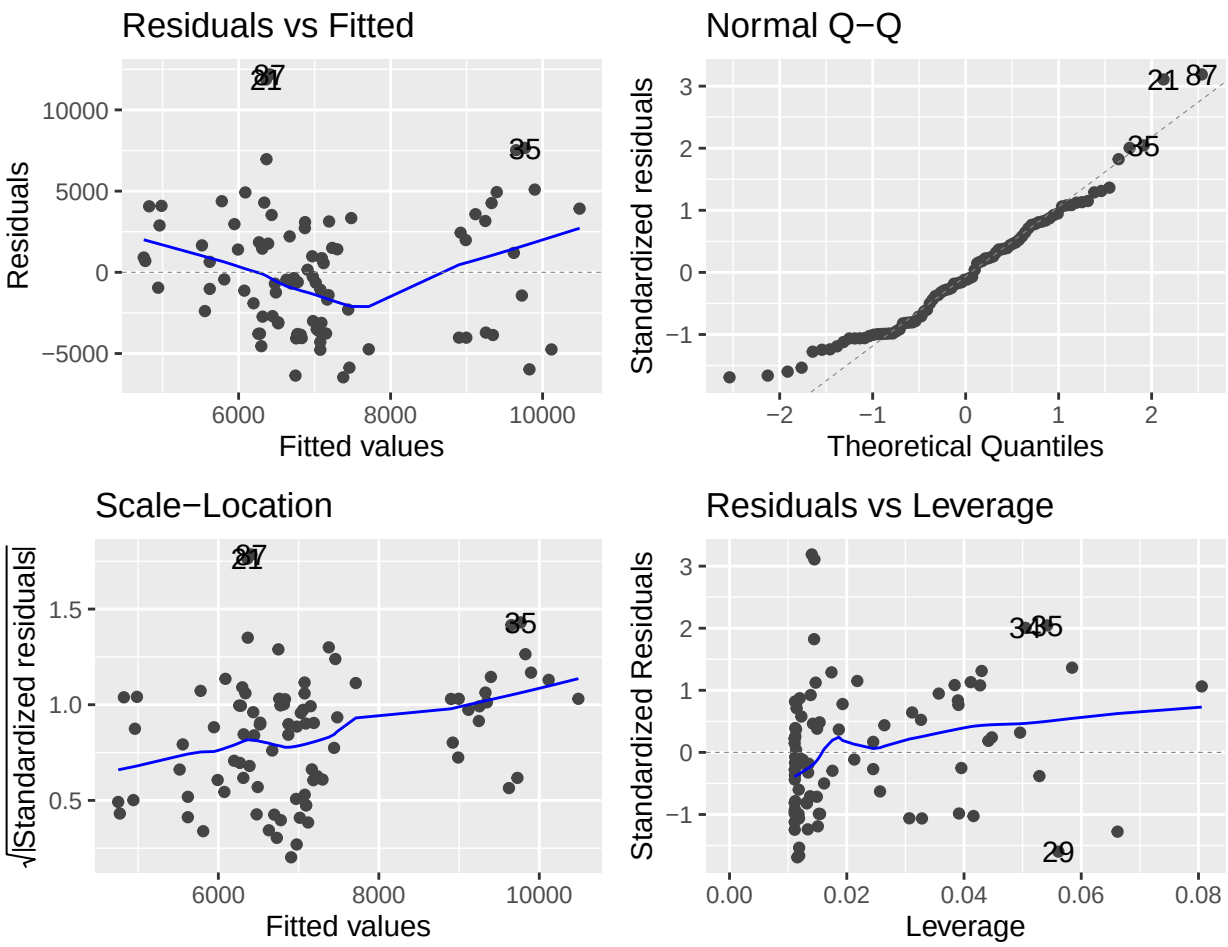


Figure D 1: Diagnostic plots on raw response

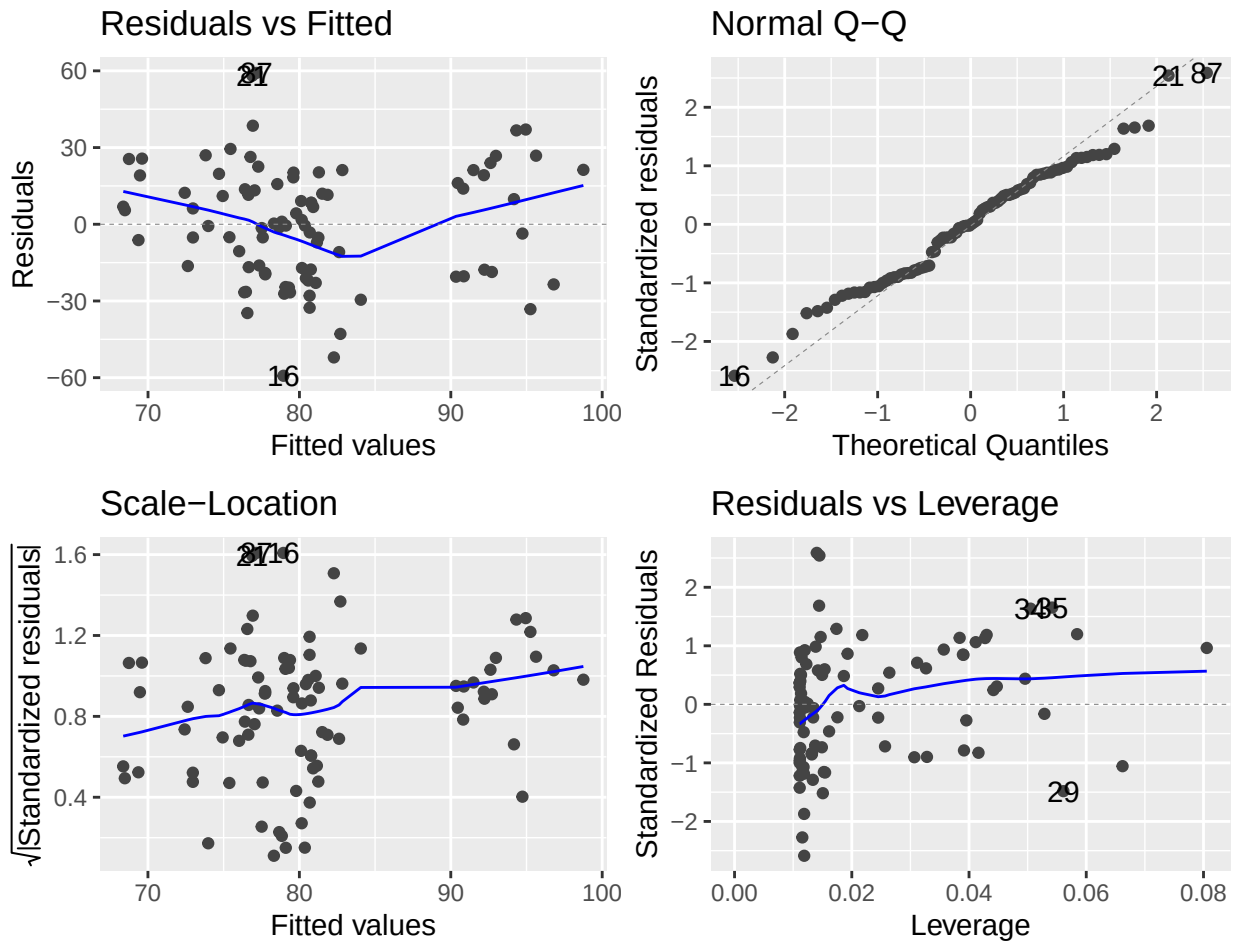


Figure D 2: Diagnostic plots after square-root transformation

Table D 1: Diagnostic tests for raw and transformed response

Test	Model	Statistic	p.value	Assumption
Shapiro-Wilk normality test	Raw	0.96	0.00	Not met
Shapiro-Wilk normality test	Transformed	0.99	0.40	Met
Non-constant Variance Score Test	Raw	1.91	0.17	Met
Non-constant Variance Score Test	Transformed	0.40	0.53	Met

Table D 2: Model fit for raw and transformed response

	biomass ~ hydro	sqrt(biomass) ~ hydro
(Intercept)	7105.21	80.84
	405.69	2.43
hydro	-3701.69	-19.61
	1110.14	6.65

	biomass ~ hydro	sqrt(biomass) ~ hydro
N	90.00	90.00
DF	88.00	88.00
AIC	1745.37	824.12
BIC	1752.87	831.62
R2	0.11	0.09
Adjusted R2	0.10	0.08
F	11.12	8.71
p-value	0.00	0.00

Nodule surface area

Nodule surface area was log-transformed to linearise relationships in upstream models in which it is an explanatory variable. Nevertheless, we also compared model fit with raw and transformed variable for the downstream path model $Nodule\ surface\ area \sim Hydrosedimentary\ stress + Surfaceconvexity$. Model fit is globally the same among both models, but the transformation is kept to linearise upstream relationships.

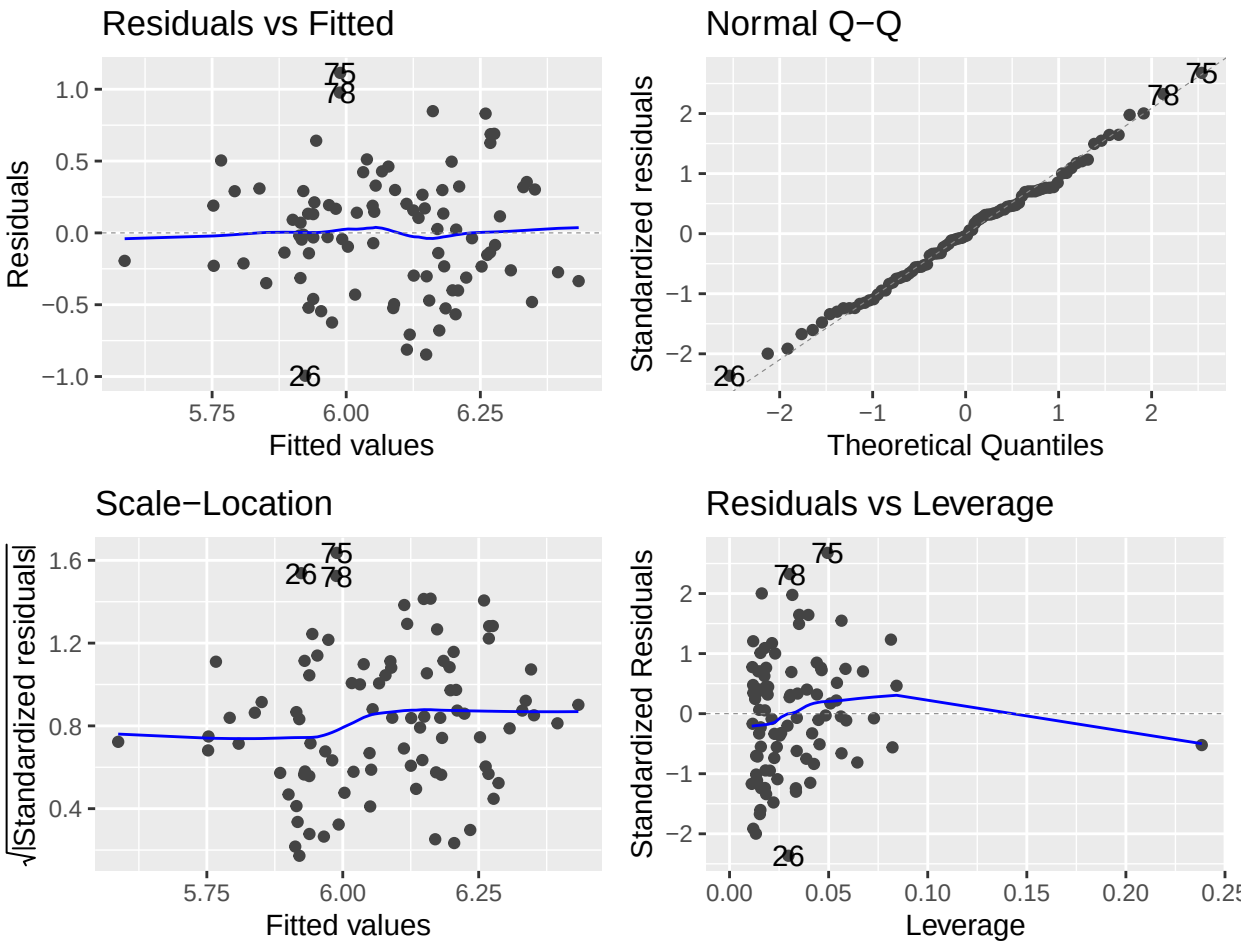


Figure D 3: Diagnostic plots on raw response

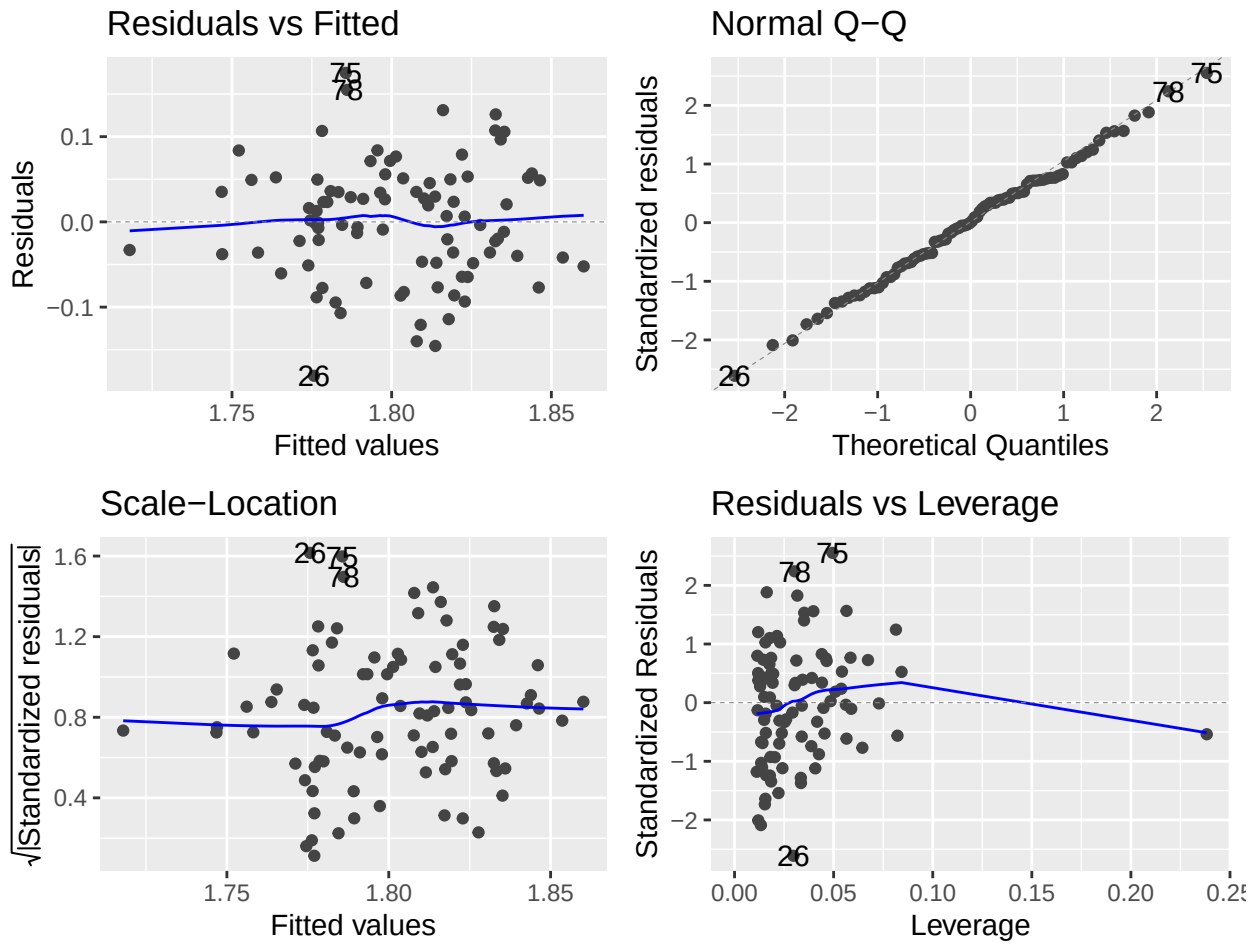


Figure D 4: Diagnostic plots after square-root transformation

Table D 3: Diagnostic tests for raw and transformed response

Test	Model	Statistic	p.value	Assumption
Shapiro-Wilk normality test	Raw	0.99	0.96	Met
Shapiro-Wilk normality test	Transformed	1.00	0.99	Met
Non-constant Variance Score Test	Raw	0.33	0.56	Met
Non-constant Variance Score Test	Transformed	0.05	0.82	Met

Table D 4: Model fit for raw and transformed response

	surf ~ hydro + convexity	log(surf) ~ hydro + convexity,
(Intercept)	6.59	1.89
	0.19	0.03
hydro	0.48	0.08
	0.14	0.02

	surf ~ hydro + convexity	log(surf) ~ hydro + convexity,
convexity	-1.38	-0.23
	0.50	0.08
N	90.00	90.00
DF	87.00	87.00
AIC	107.12	-217.77
BIC	117.12	-207.77
R2	0.13	0.14
Adjusted R2	0.11	0.12
F	6.77	6.80
p-value	0.00	0.00

Habitat geometry

Diagnostic plots and tests for the path model $Convexity \sim Hydrosedimentary\ stress$ on the raw and transformed response variable. Even after transformations, linear model assumptions were not met. Therefore, an alternative generalized linear model (GLM) with a Gamma distribution and inverse link was fitted to better capture the nonlinear, hyperbolic and asymptotic effects of hydrosedimentary stress. To assess whether the Gamma model provided a meaningfully different predictive outcome than the Gaussian model, we compared the predicted values from both models using paired predictions on the same data. Because the mean difference between predictions was not statistically significant, and predictive accuracy (RMSE/MAE) was similar between models, the Gaussian model was retained for the pSEM to allow comparison of standardized path coefficients (Supplementary Information).

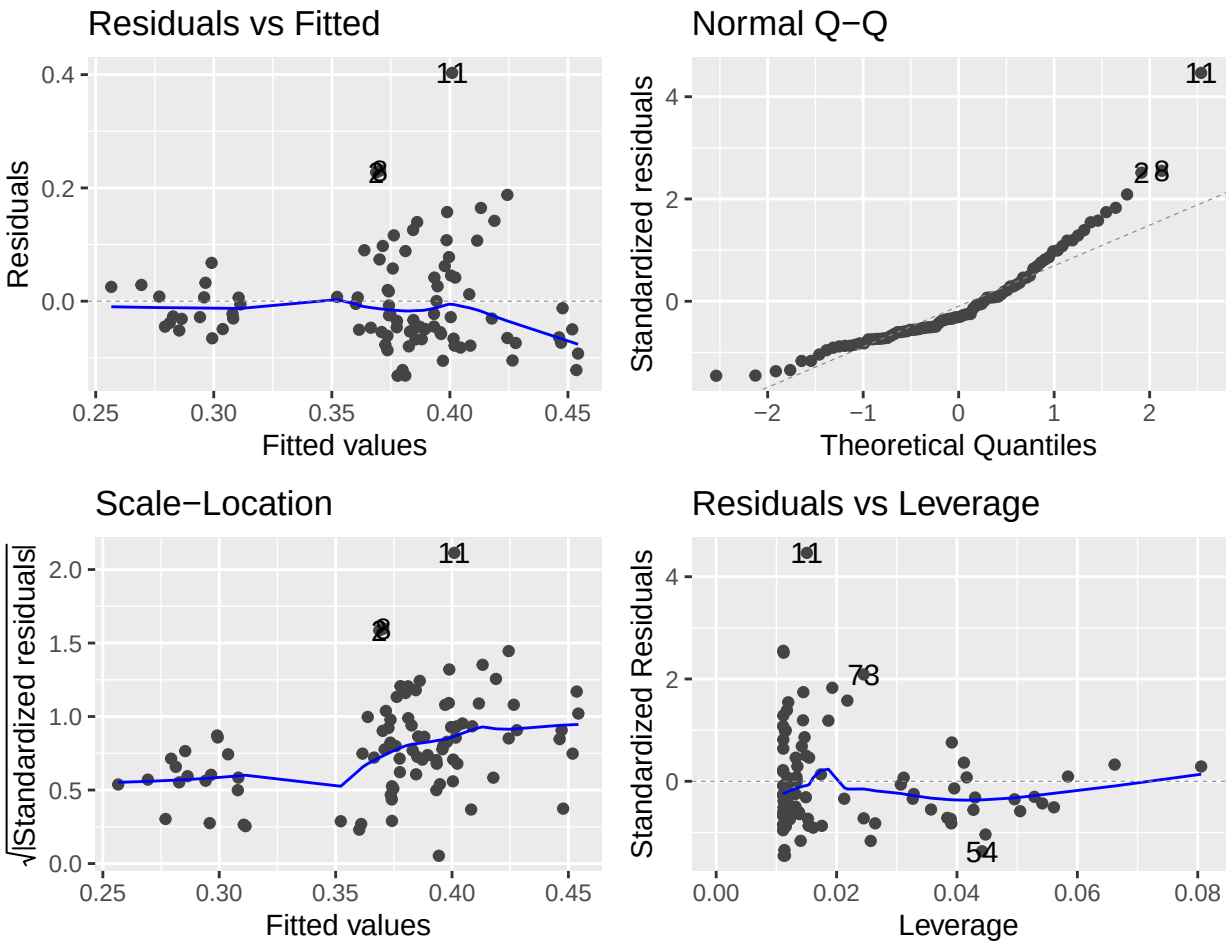


Figure D 5: Diagnostic plots on raw response

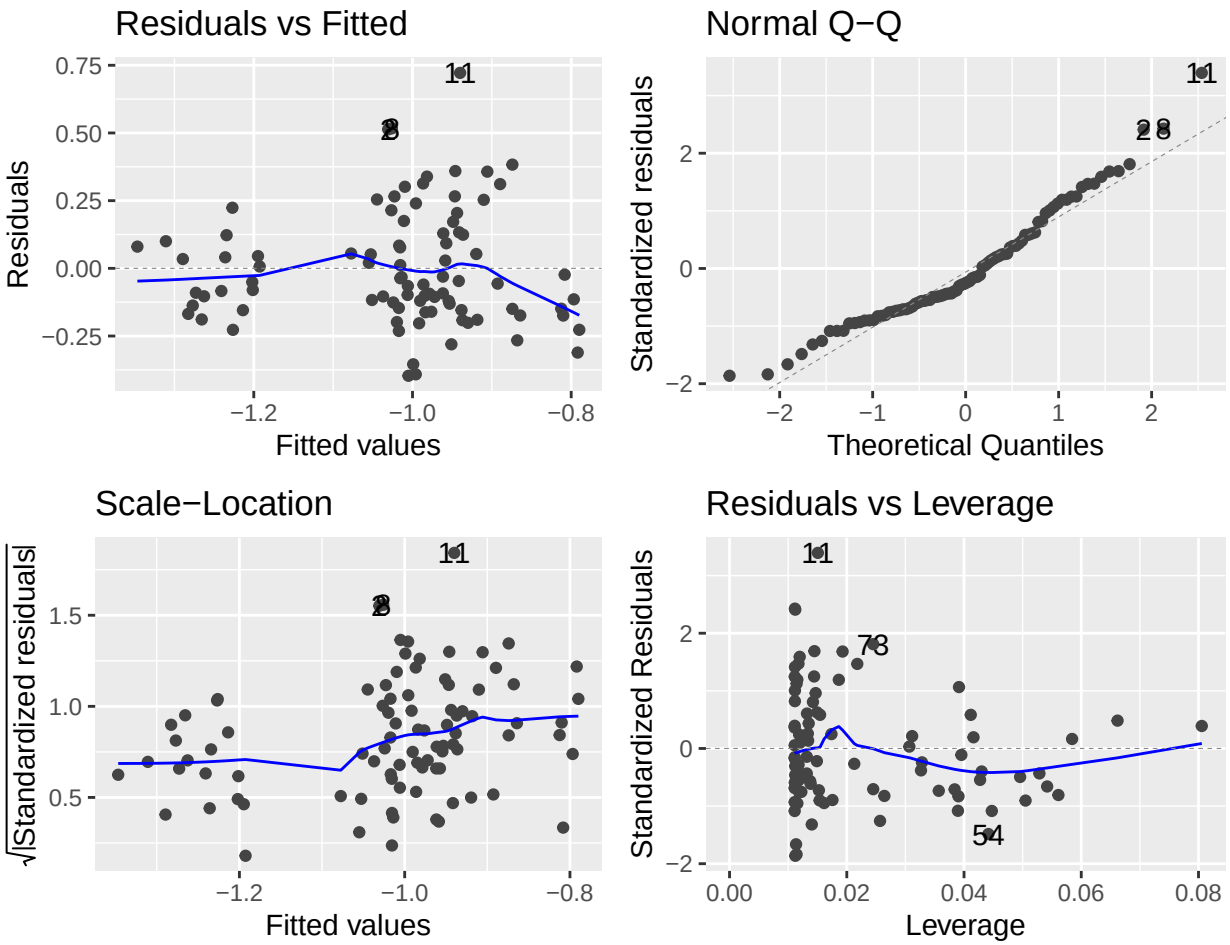


Figure D 6: Diagnostic plots after log transformation

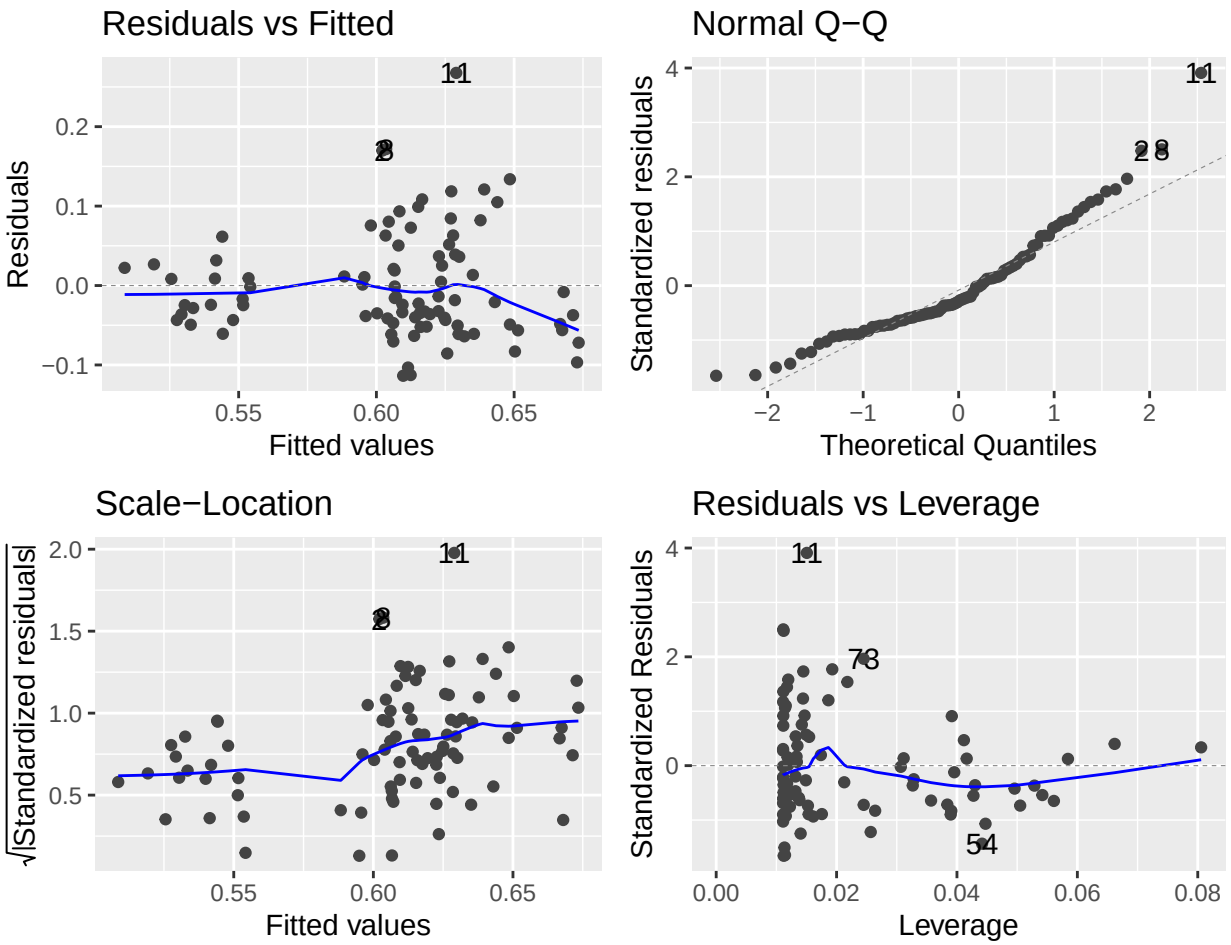


Figure D 7: Diagnostic plots after square-root transformation

Table D 5: Diagnostic tests for raw and transformed response

Test	Model	Statistic	p.value	Assumption
Shapiro-Wilk normality test	Raw	0.88	0.00	Not met
Shapiro-Wilk normality test	Log-transformed	0.95	0.00	Not met
Shapiro-Wilk normality test	Square root-transformed	0.92	0.00	Not met
Non-constant Variance Score Test	Raw	7.64	0.01	Not met
Non-constant Variance Score Test	Log-transformed	3.89	0.05	Not met
Non-constant Variance Score Test	Square root-transformed	5.80	0.02	Not met

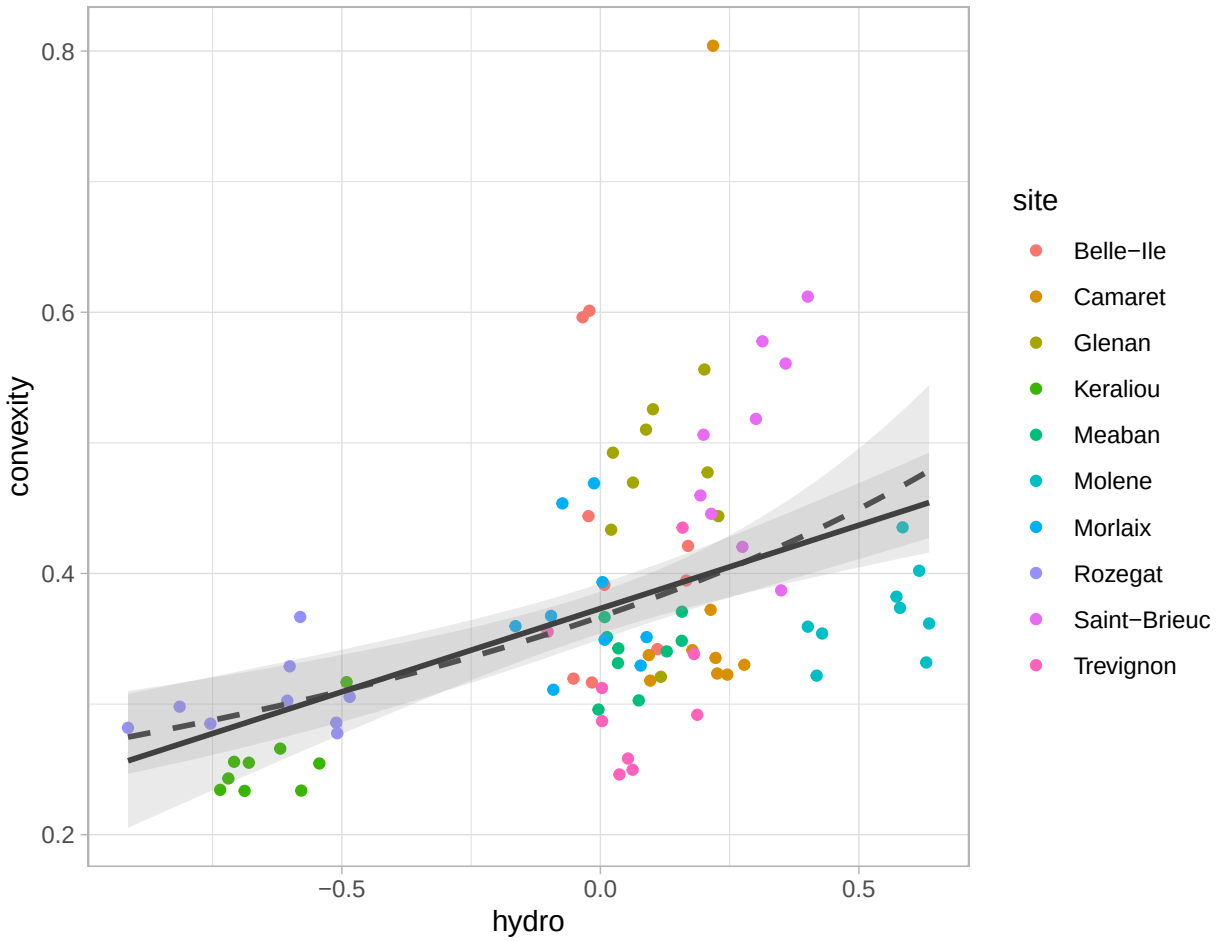


Figure D 8: Comparison between Gamma glm (dashed line) and gaussian lm (full line)

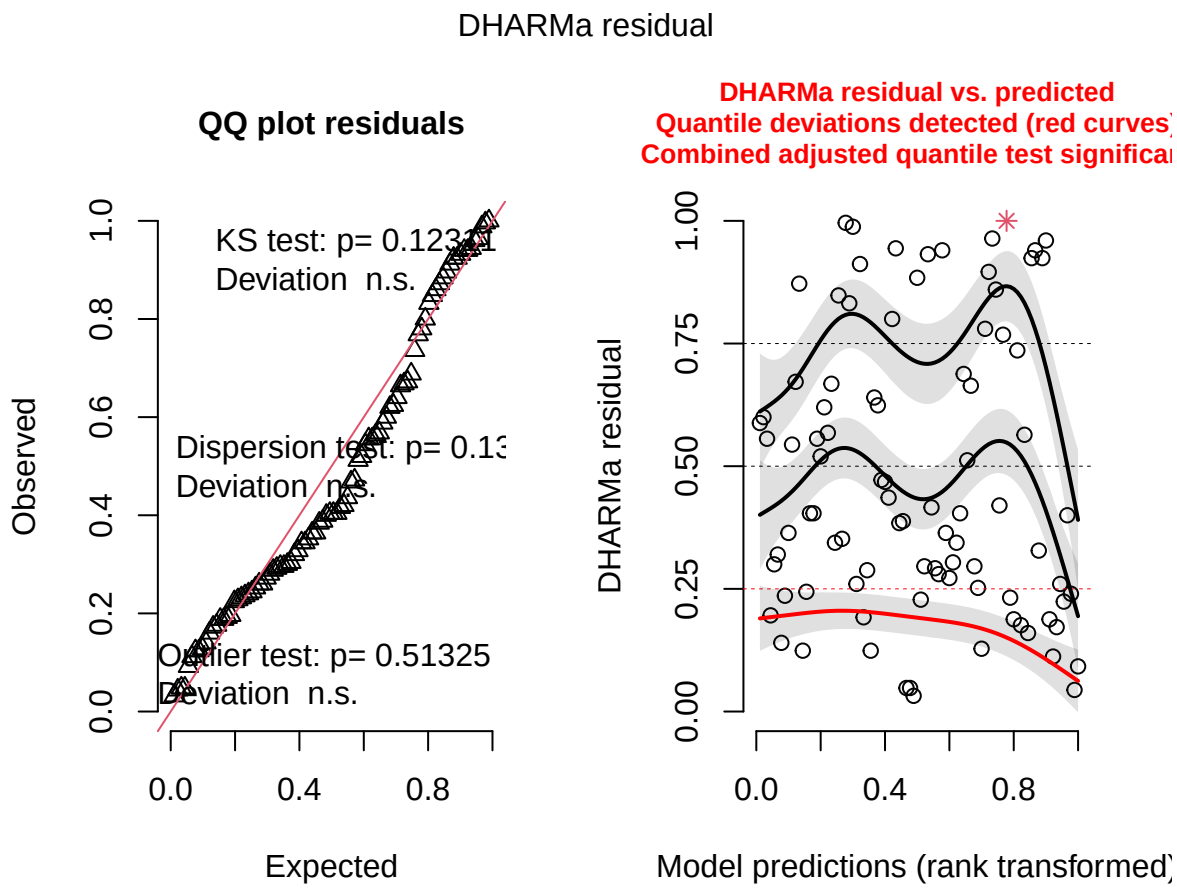


Figure D 9: Diagnostic plots of Gamma glm with inverse link

Table D 6: Model fit of linear models with raw and transformed response and of generalised linear model with gamma distributed errors and inverse link

	convexity ~ hydro	log(convexity) ~ hydro	sqrt(convexity) ~ hydro	convexity ~ hydro, family = Gamma(link = "inverse")
(Intercept)	0.37	-1.02	0.61	2.73
	0.01	0.02	0.01	0.07
hydro	0.13	0.36	0.11	-1
	0.03	0.06	0.02	0.2
N	90.00	90.00	90.00	90
DF	88.00	88.00	88.00	88
AIC	-172.15	-17.88	-221.92	-193.43
BIC	-164.65	-10.38	-214.42	-185.93
R2	0.21	0.28	0.25	
Adjusted R2	0.20	0.27	0.24	

	convexity ~ hydro	log(convexity) ~ hydro	sqrt(convexity) ~ hydro	convexity ~ hydro, family = Gamma(link = "inverse")
F	23.65	33.81	28.58	
p-value	0.00	0.00	0.00	

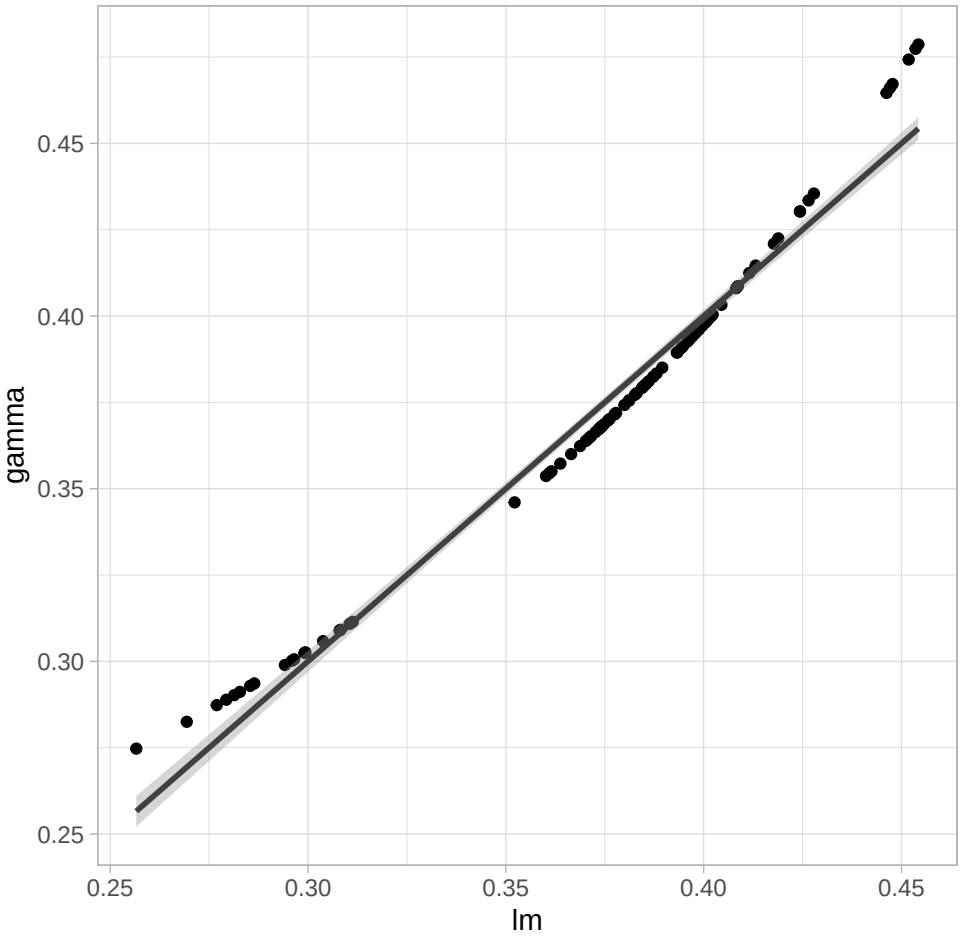


Figure D 10: Predicted values from Gaussian lm and Gamma glm

Table D 7: Differences between predicted values from lm and glm

Test	Statistic	df	p.value
Paired t-test	0	89	1.000
Wilcoxon signed rank test with continuity correction	2422	89	0.132

Table D 8: Differences in predictive power from lm and glm

Statistic	LM	GLM	Diff
MAE	0.068	0.069	0.001
RMSE	0.090	0.092	0.002

Habitat diversity

Diagnostic plots and tests for the path model $Nodule\,trait\,dispersion \sim Hydrosedimentary\,stress$ with raw and first degree polynomial response variable.

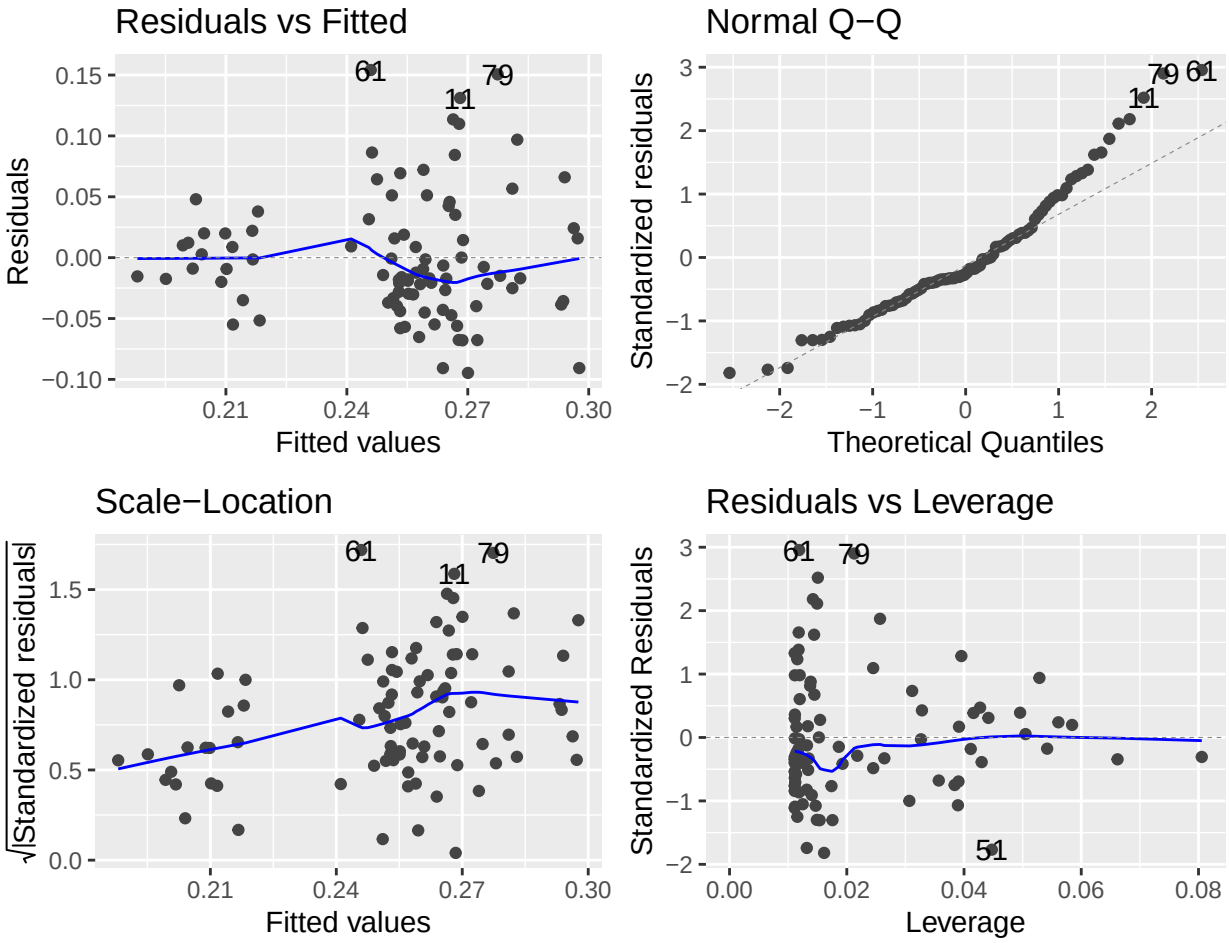


Figure D 11: Diagnostic plots on raw response

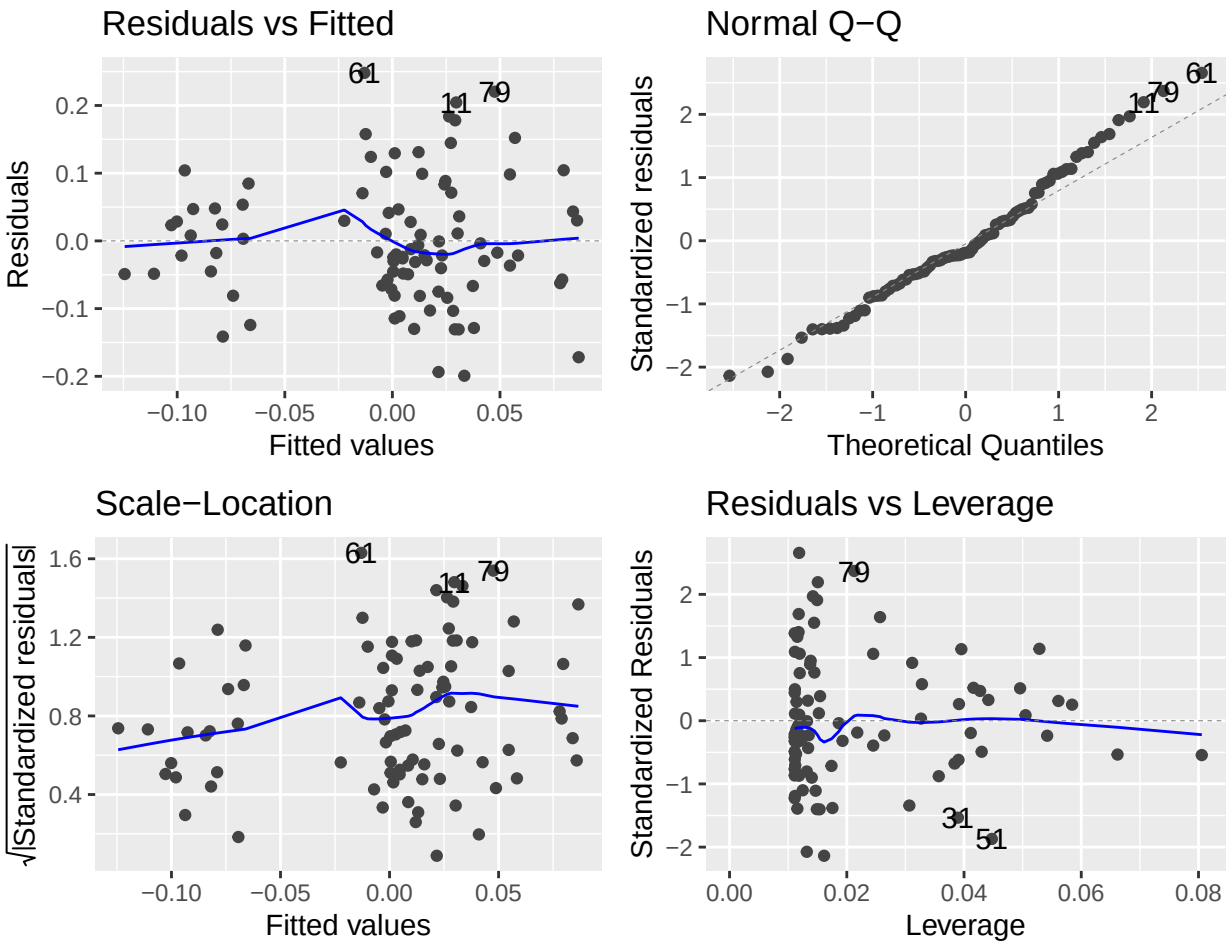


Figure D 12: Diagnostic plots first dregree polynomial

Table D 9: Diagnostic tests for raw and transformed response

Test	Model	Statistic	p.value	Assumption
Shapiro-Wilk normality test	Raw	0.95	0.00	Not met
Shapiro-Wilk normality test	First-degree polynomial	0.98	0.26	Met
Non-constant Variance Score Test	Raw	6.78	0.01	Not met
Non-constant Variance Score Test	First-degree polynomial	3.40	0.07	Met

Table D 10: Model fit of linear models with raw and transformed response

	mfdis ~ hydro	mfdis_1 ~ hydro
(Intercept)	0.25	0.00
	0.01	0.01
hydro	0.07	0.14
	0.02	0.03

	mfdis ~ hydro	mfdis_1 ~ hydro
N	90.00	90.00
DF	88.00	88.00
AIC	-271.22	-166.27
BIC	-263.72	-158.77
R2	0.20	0.22
Adjusted R2	0.19	0.21
F	21.86	25.24
p-value	0.00	0.00

Functional diversity

Functional identity

Diagnostic plots and tests for the path model $Functional\,identity \sim Hydrosedimentary\,stress + Habitat\ amount\ +\ Habitat\ geometry\ +\ Habitat\ diversity.$ All models respected linear model assumptions. Model fit among models containing raw values for nodule morphological trait dispersion (mfdis), only the first order polynomial of nodule trait dispersion, and both polynomials are shown.

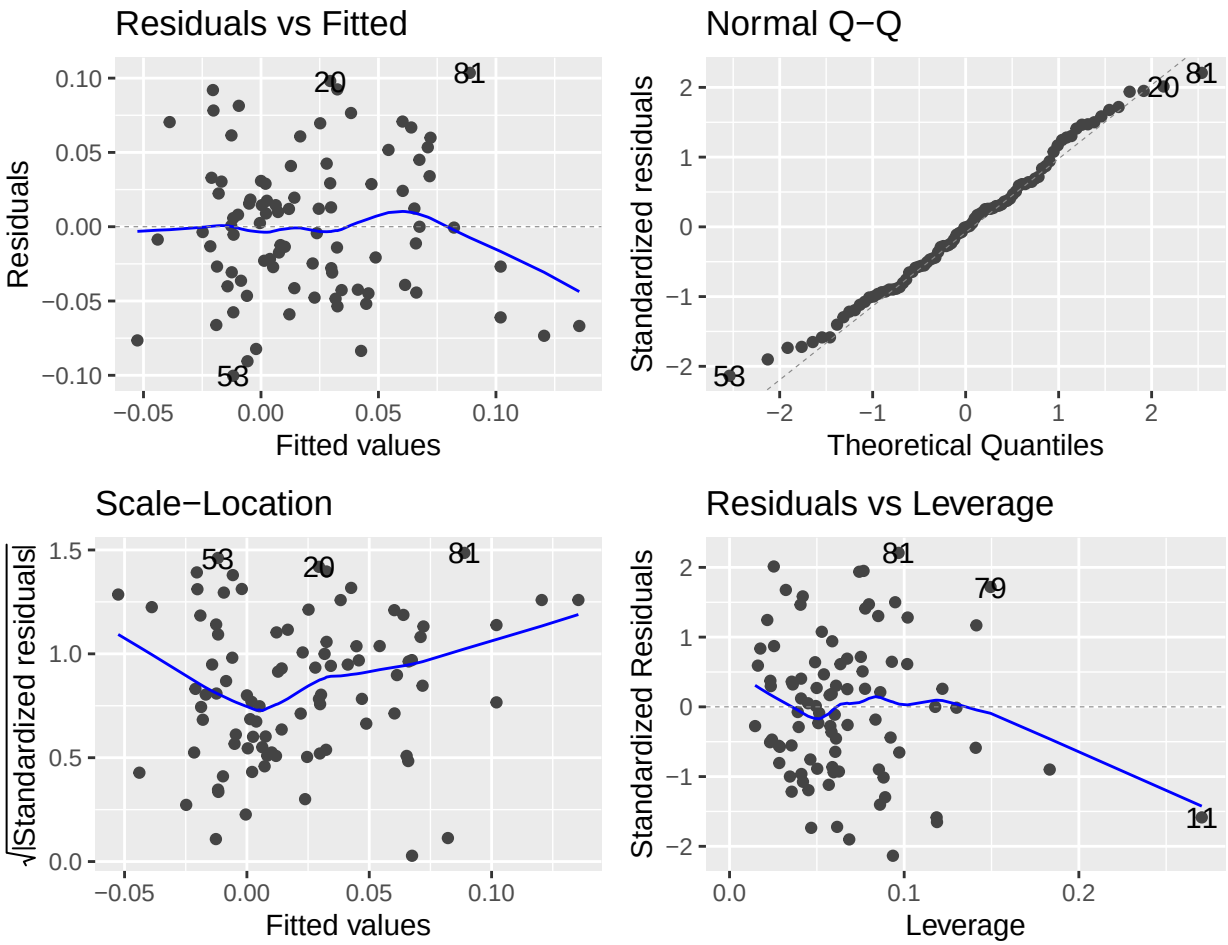


Figure D 13: Diagnostic plots

Table D 11: Diagnostic tests for raw and transformed response

Test	Model	Statistic	p.value	Assumption
Shapiro-Wilk normality test	fide_Dim1 ~ hydro + mfdis + convexity + log(surf) + sqrt(biomass)	0.99	0.43	Met
Shapiro-Wilk normality test	fide_Dim1 ~ hydro + mfdis_1 + convexity + log(surf) + sqrt(biomass)	0.99	0.48	Met
Shapiro-Wilk normality test	fide_Dim1 ~ hydro + mfdis_1 + mfdis_2 + convexity + log(surf) + sqrt(biomass)	0.98	0.38	Met
Non-constant Variance Score Test	fide_Dim1 ~ hydro + mfdis + convexity + log(surf) + sqrt(biomass)	0.52	0.47	Met
Non-constant Variance Score Test	fide_Dim1 ~ hydro + mfdis_1 + convexity + log(surf) + sqrt(biomass)	0.57	0.45	Met

Test	Model	Statistic	p.value	Assumption
Non-constant Variance Score Test	fide_Dim1 ~ hydro + mfdis_1 + mfdis_2 + convexity + log(surf) + sqrt(biomass)	0.52	0.47	Met

Table D 12: Model fit of linear models with raw and polynomial terms of nodule trait dispersion

	Raw response	1st order polynomial predictors	1st and 2nd order polynomial predictors
(Intercept)	-0.14	-0.17	-0.18
	0.15	0.15	0.15
hydro	-0.06	-0.06	-0.06
	0.02	0.02	0.02
mfdis	-0.12		
	0.11		
convexity	0.39	0.38	0.39
	0.06	0.06	0.06
log(surf)	0.01	0.01	0.01
	0.08	0.08	0.08
sqrt(biomass)	0	0	0
	0	0	0
mfdis–1		-0.06	-0.06
		0.06	0.06
mfdis–2			-0.03
			0.05
N	90	90	90
DF	84	84	83
AIC	-278.45	-278.28	-276.66
BIC	-260.95	-260.78	-256.66
R2	0.38	0.37	0.38
Adjusted R2	0.34	0.34	0.33
F	10.11	10.06	8.38
p-value	0	0	0

Functional evenness

Diagnostic plots and tests for the path model $Functional\,identity \sim Hydrosedimentary\,stress + Habitat\,amount\ +\ Habitat\,geometry\ +\ Habitat\,diversity.$

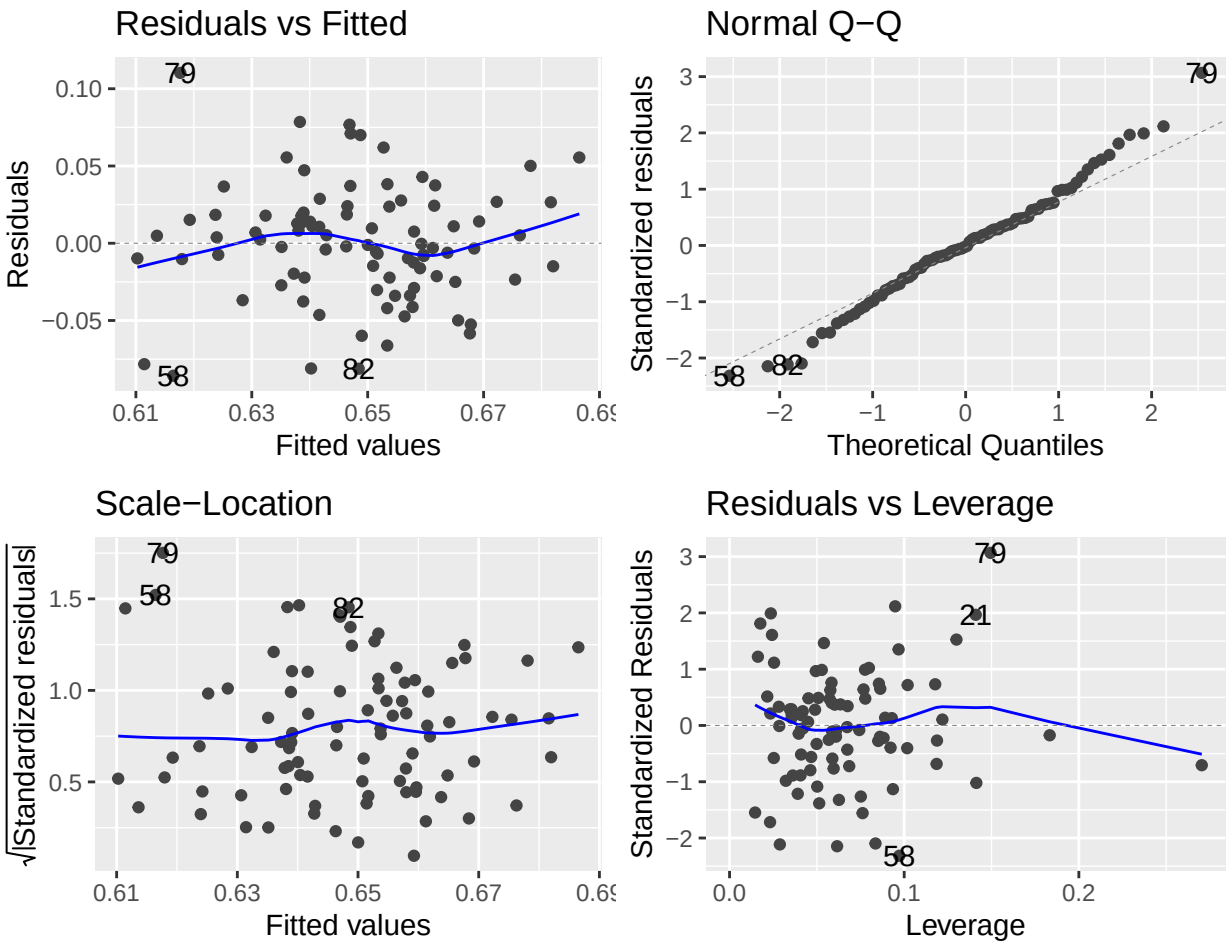


Figure D 14: Diagnostic plots on raw response

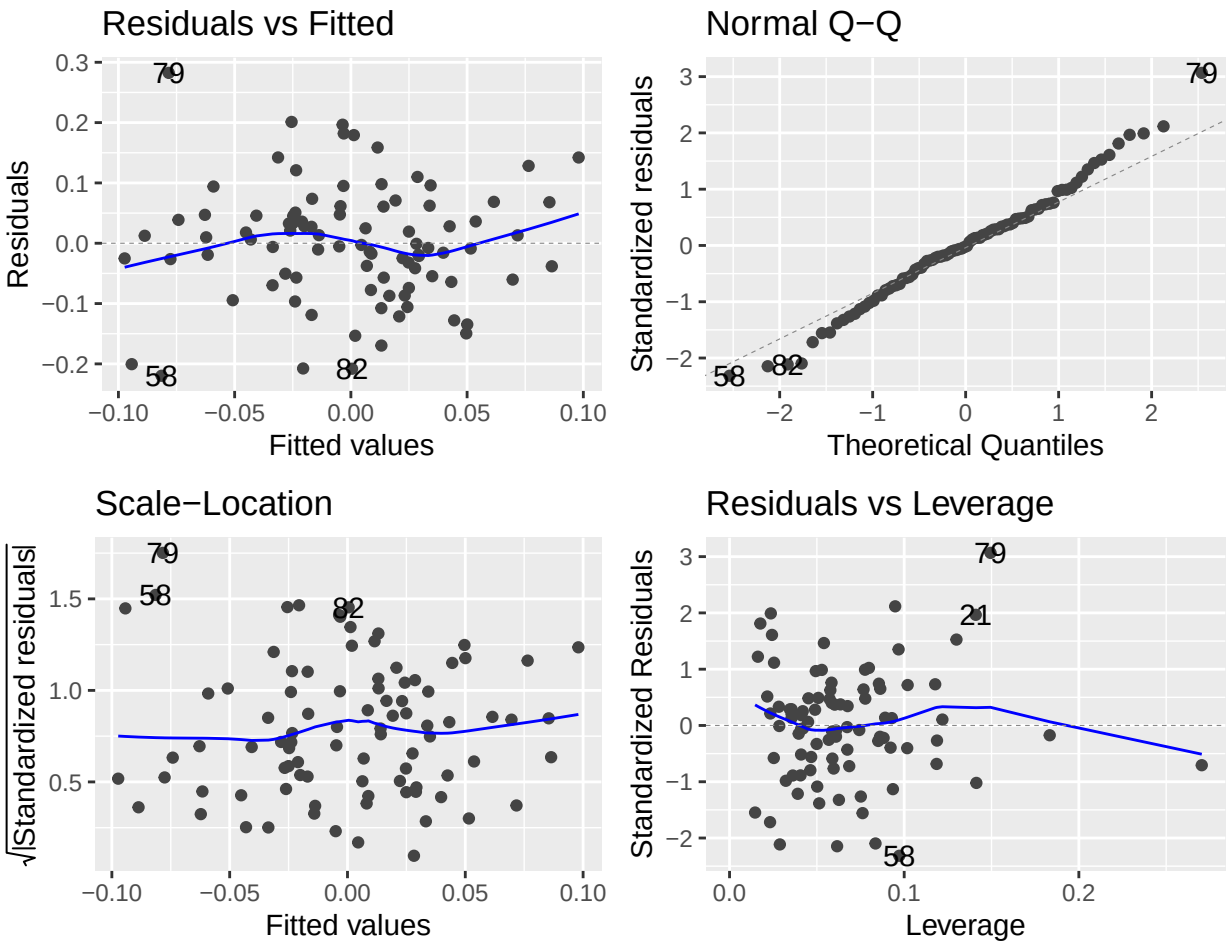


Figure D 15: Diagnostic plots first dregree polynomial

Table D 13: Diagnostic tests for raw and transformed response

Test	Model	Statistic	p.value	Assumption
Shapiro-Wilk normality test	feve ~ hydro + mfdis + convexity + sqrt(biomass) + log(surf)	0.99	0.67	Met
Shapiro-Wilk normality test	feve_1 ~ hydro + mfdis + convexity + sqrt(biomass) + log(surf)	0.99	0.67	Met
Shapiro-Wilk normality test	feve_1 ~ hydro + mfdis_1 + convexity + sqrt(biomass) + log(surf)	0.99	0.63	Met
Shapiro-Wilk normality test	feve_1 ~ hydro + mfdis_1 + mfdis_2 + convexity + sqrt(biomass) + log(surf)	0.99	0.66	Met
Non-constant Variance Score Test	feve ~ hydro + mfdis + convexity + sqrt(biomass) + log(surf)	2.91	0.09	Met
Non-constant Variance Score Test	feve_1 ~ hydro + mfdis + convexity + sqrt(biomass) + log(surf)	2.91	0.09	Met

Test	Model	Statistic	p.value	Assumption
Non-constant Variance Score Test	feve_1 ~ hydro + mfdis_1 + convexity + sqrt(biomass) + log(surf)	2.97	0.08	Met
Non-constant Variance Score Test	feve_1 ~ hydro + mfdis_1 + mfdis_2 + convexity + sqrt(biomass) + log(surf)	1.83	0.18	Met

Table D 14: Model fit of linear models with raw and transformed response and polynomial terms for nodule trait dispersion

	Raw response	1st order polynomial response	1st order polynomial predictors	1st and 2nd order polynomial predictors
(Intercept)	0.27	-0.96	-1.02	-1.02
	0.12	0.3	0.3	0.31
hydro	-0.05	-0.12	-0.11	-0.11
	0.01	0.04	0.04	0.04
mfdis	-0.08	-0.2		
	0.08	0.22		
convexity	0.15	0.38	0.39	0.38
	0.05	0.13	0.13	0.13
sqrt(biomass)	0	0	0	0
	0	0	0	0
log(surf)	0.2	0.51	0.51	0.51
	0.07	0.17	0.17	0.17
mfdis–1			-0.13	-0.14
			0.12	0.12
mfdis–2				0.09
				0.1
N	90	90	90	90
DF	84	84	84	83
AIC	-320.82	-151.52	-151.91	-150.82
BIC	-303.32	-134.02	-134.42	-130.82
R2	0.16	0.16	0.17	0.17
Adjusted R2	0.11	0.11	0.12	0.11
F	3.26	3.26	3.34	2.92
p-value	0.01	0.01	0.01	0.01

Community size

Total polychaete density

Diagnostic plots and tests for the path model $Polychaete\ density \sim Hydrosedimentary\ stress + Habitat\ amount\ +\ Habitat\ geometry\ +\ Habitat\ diversity\ +\ Trait\ diversity.$

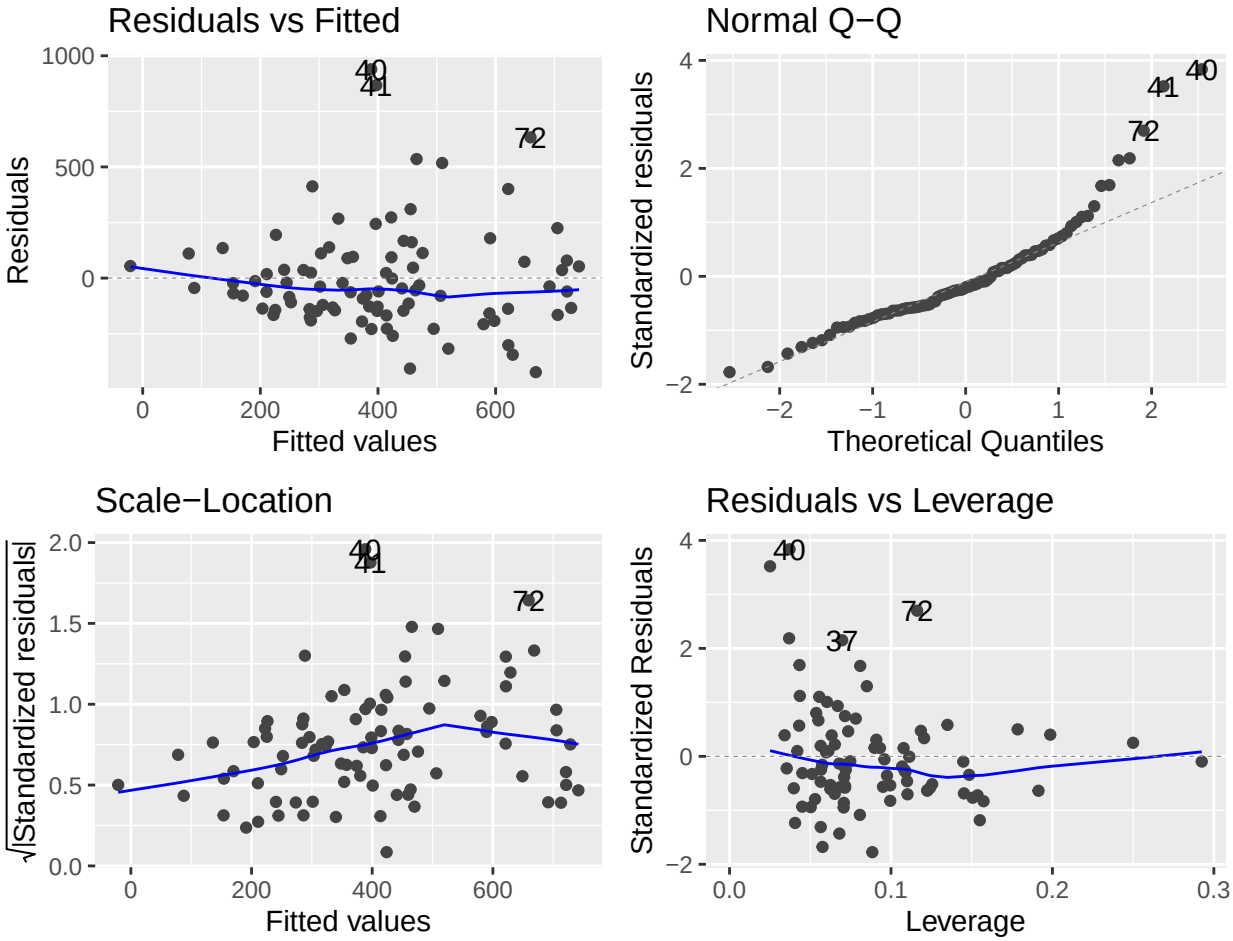


Figure D 16: Diagnostic plots on raw response

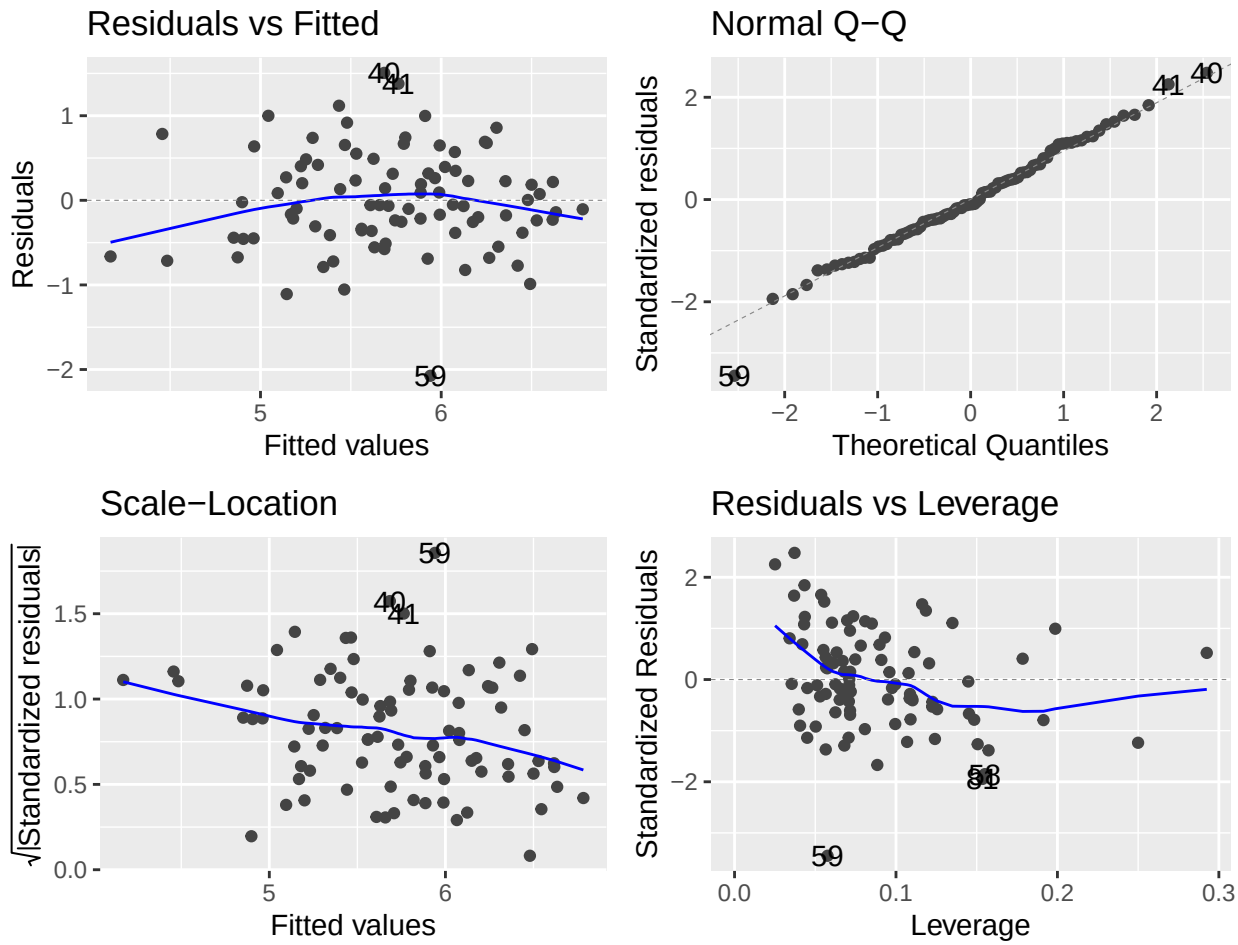


Figure D 17: Diagnostic plots log-transformed response

Table D 15: Diagnostic tests for raw and transformed response

Test	Model	Statistic	p.value	Assumption
Shapiro-Wilk normality test	dens ~ hydro + mfdis + convexity + sqrt(biomass) + log(surf) + fide_Dim1 + feve	0.88	0.00	Not met
Shapiro-Wilk normality test	log(dens) ~ hydro + mfdis + convexity + sqrt(biomass) + log(surf) + fide_Dim1 + feve	0.99	0.49	Met
Shapiro-Wilk normality test	log(dens) ~ hydro + mfdis_1 + convexity + sqrt(biomass) + log(surf) + fide_Dim1 + feve_1	0.99	0.72	Met
Shapiro-Wilk normality test	log(dens) ~ hydro + mfdis_1 + mfdis_2 + convexity + sqrt(biomass) + log(surf) + fide_Dim1 + feve_1 + feve_2	0.98	0.18	Met
Non-constant Variance Score Test	dens ~ hydro + mfdis + convexity + sqrt(biomass) + log(surf) + fide_Dim1 + feve	4.29	0.04	Not met

Test	Model	Statistic	p.value	Assumption
Non-constant Variance Score Test	log(dens) ~ hydro + mfdis + convexity + sqrt(biomass) + log(surf) + fide_Dim1 + feve	0.69	0.41	Met
Non-constant Variance Score Test	log(dens) ~ hydro + mfdis_1 + convexity + sqrt(biomass) + log(surf) + fide_Dim1 + feve_1	1.27	0.26	Met
Non-constant Variance Score Test	log(dens) ~ hydro + mfdis_1 + mfdis_2 + convexity + sqrt(biomass) + log(surf) + fide_Dim1 + feve_1 + feve_2	0.21	0.65	Met

Table D 16: Model fit of linear models with raw and transformed response and raw and polynomial terms for functional eveness and nodule trait dispersion

	Raw response	Transformed response	1st order polynomial predictors	1st and 2nd order polynomial predictors
(Intercept)	906.8	8.8	7.3	7.44
	797.92	1.98	2.06	1.93
hydro	-341.84	-0.77	-0.75	-0.78
	103.59	0.26	0.26	0.25
mfdis	-1587.81	-6.9		
	549.11	1.36		
convexity	477.26	1.67	1.53	1.41
	391.1	0.97	0.99	0.93
	0.9	0	0	0
sqrt(biomass)				
	1.33	0	0	0
log(surf)	-198.47	-1.26	-1.27	-1.36
	442.39	1.1	1.12	1.05
fide–Dim1	-1002.52	-3.68	-3.52	-3.64
	586.59	1.46	1.49	1.4
feve	43.49	0.29		
	742.28	1.84		
mfdis–1			-3.56	-3.06
			0.78	0.74
feve–1			0.05	0.33
			0.74	0.69
mfdis–2				-1.71
				0.61
feve–2				-1.54

	Raw response	Transformed response	1st order polynomial predictors	1st and 2nd order polynomial predictors
				0.62
N	90	90	90	90
DF	82	82	82	80
AIC	1258.56	178.92	182.89	172.06
BIC	1281.05	201.42	205.39	199.56
R2	0.33	0.45	0.43	0.52
Adjusted R2	0.28	0.41	0.38	0.46
F	5.85	9.73	8.8	9.47
p-value	0	0	0	0

Polychaete richness

Diagnostic plots and tests for the path model $Polychaete\,richness \sim Hydrosedimentary\,stress + Habitat\ amount\ +\ Habitat\ geometry\ +\ Habitat\ diversity\ +\ Trait\ diversity + Polychaete\,density.$

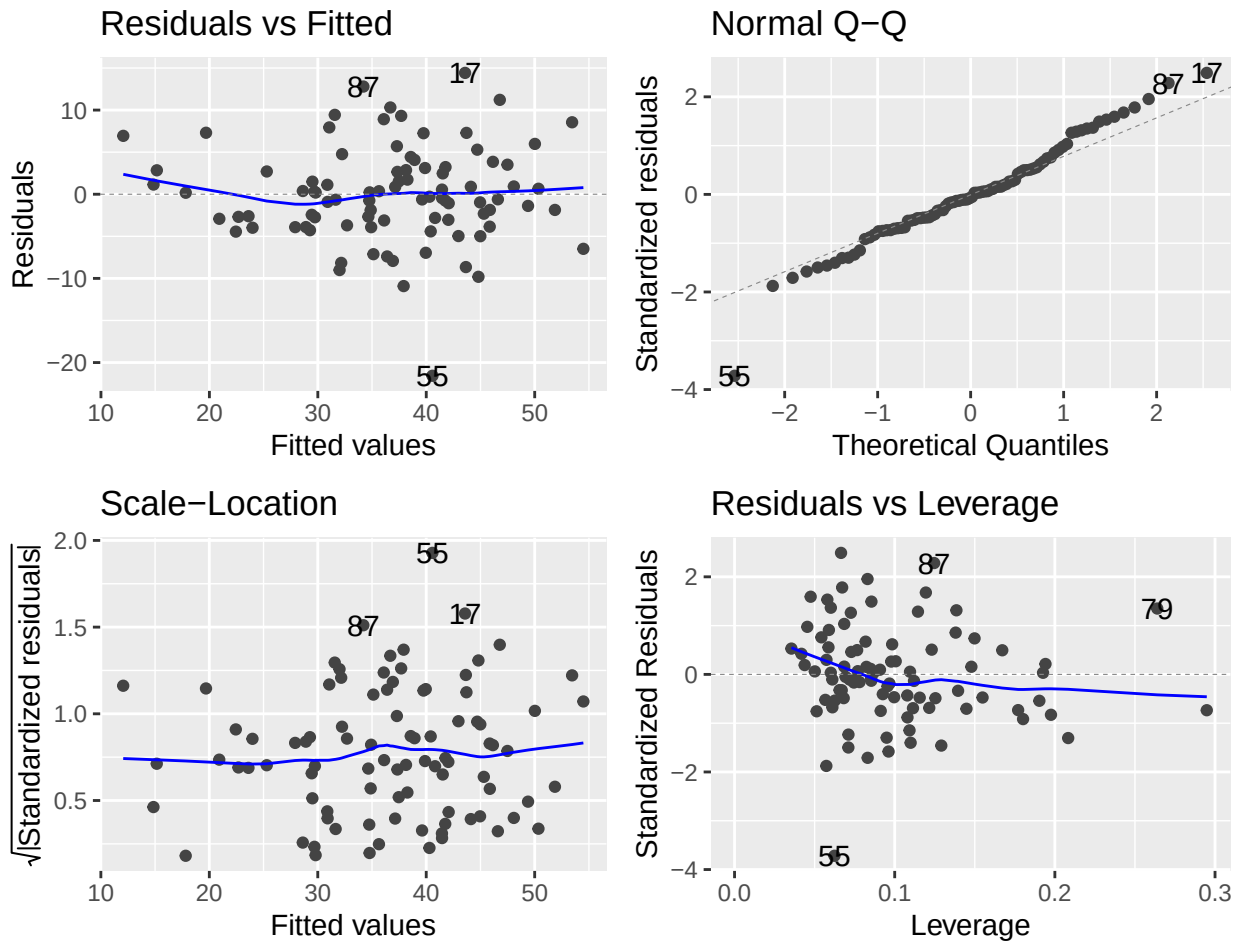


Figure D 18: Diagnostic plots on raw response

Table D 17: Diagnostic tests for models with raw and polynomial terms for explanatory variables

Test	Model	Statistic	p.value	Assumption
Shapiro-Wilk normality test	sp_richn ~ hydro + mfdis + convexity + sqrt(biomass) + log(surf) + fide_Dim1 + feve + log(dens)	0.97	0.07	Met
Shapiro-Wilk normality test	sp_richn ~ hydro + mfdis_1 + convexity + sqrt(biomass) + log(surf) + fide_Dim1 + feve_1 + log(dens)	0.97	0.07	Met
Shapiro-Wilk normality test	sp_richn ~ hydro + mfdis_1 + mfdis_2 + convexity + sqrt(biomass) + log(surf) + fide_Dim1 + feve_1 + feve_2 + log(dens)	0.98	0.09	Met
Non-constant Variance Score Test	sp_richn ~ hydro + mfdis + convexity + sqrt(biomass) + log(surf) + fide_Dim1 + feve + log(dens)	1.66	0.20	Met
Non-constant Variance Score Test	sp_richn ~ hydro + mfdis_1 + convexity + sqrt(biomass) + log(surf) + fide_Dim1 + feve_1 + log(dens)	1.61	0.20	Met

Test	Model	Statistic	p.value	Assumption
Non-constant Variance Score Test	sp_richn ~ hydro + mfdis_1 + mfdis_2 + convexity + sqrt(biomass) + log(surf) + fide_Dim1 + feve_1 + feve_2 + log(dens)	1.83	0.18	Met

Table D 18: Model fit of linear models with raw and polynomial terms for explanatory variables

	Raw response	1st order polynomial predictors	1st and 2nd order polynomial predictors
(Intercept)	-40.14	-40.41	-39.27
	21.34	20.97	21.1
hydro	2.24	2.2	3.06
	2.62	2.62	2.69
mfdis	4.86		
	15.1		
convexity	-3.52	-2.9	-4.82
	9.56	9.52	9.51
sqrt(biomass)	-0.02	-0.02	-0.01
	0.03	0.03	0.03
log(surf)	6.93	6.95	6.94
	10.71	10.72	10.66
fide–Dim1	5.25	4.68	7.33
	14.62	14.55	14.67
feve	-2.76		
	17.83		
log(dens)	11.76	11.67	11.49
	1.07	1.04	1.12
mfdis–1		1.32	2.66
		8.25	8.21
feve–1		-1.1	-1.69
		6.97	6.98
mfdis–2			5.38
			6.42
feve–2			-10.28
			6.47
N	90	90	90
DF	81	81	79
AIC	588.2	588.29	588.48
BIC	613.2	613.28	618.48
R2	0.71	0.71	0.72

	Raw response	1st order polynomial predictors	1st and 2nd order polynomial predictors
Adjusted R2	0.68	0.68	0.68
F	24.37	24.33	20.15
p-value	0	0	0

SEM fit and selection

Global pSEM goodness of fit was assessed through Fisher's C statistic, an Akaike Information Criterion based on directed separation tests (AICd-sep; (Shipley 2013)), as well as the generalised maximum-likelihood chi-square statistic (□2ML) and full-model AICML statistic for piecewise SEMs (Shipley and Douma 2020). Initial models would only contain the hypothetical causal paths, and missing paths or correlated errors were incorporated only when identified as significant by directed separation tests and deemed ecologically relevant. Because both approaches yielded equivalent predictions and model fit, we retained the Gaussian pSEM 3 including polynomial terms for the main analyses in order to facilitate comparison of standardized path coefficients across model components while still taking into account significant non-linear relationships involving nodule trait dispersion (habitat diversity) and functional evenness.

pSEM 1

pSEM fit for the model without polynomial terms:

sp_richn ~ hydro + mfdis + convexity + sqrt(biomass) + log(surf) + fide_Dim1 + feve + log(dens),

log(dens) ~ hydro + mfdis + convexity + sqrt(biomass) + log(surf) + fide_Dim1 + feve

feve ~ hydro + mfdis + convexity + sqrt(biomass) + log(surf),

fide_Dim1 ~ hydro + mfdis + convexity + log(surf) + sqrt(biomass)

mfdis ~ hydro

convexity ~ hydro

sqrt(biomass) ~ hydro

log(surf) ~ hydro + convexity

log(surf) ~~ sqrt(biomass)

feve ~~ fide_Dim1

mfdis ~~ convexity

Table D 19: Tests of directed separation

Independ.Claim	Test.Type	DF	Crit.Value	P.Value
sqrt(biomass) ~ mfdis + …	coef	87	-1.3544	0.1791
log(surf) ~ mfdis + …	coef	86	0.2231	0.8240
sqrt(biomass) ~ convexity + …	coef	87	-0.2273	0.8207

Table D 20: Global goodness-of-fit

Chisq	df	P.Value	Fisher.C	df	P.Value
1.981	3	0.576	4.222	6	0.647

Table D 21: Model coefficients

Response	Predictor	Estimate	Std.Error	DF	Crit.Value	P.Value	Std.Estimate	
sp_richn	hydro	2.2438	2.6211	81	0.8560	0.3945	0.0782	
sp_richn	mfdis	4.8625	15.1017	81	0.3220	0.7483	0.0269	
sp_richn	convexity	-3.5239	9.56	81	-0.3686	0.7134	-0.0340	
sp_richn	sqrt(biomass)	-0.0155	0.0322	81	-0.4807	0.6320	-0.0352	
sp_richn	log(surf)	6.9252	10.7081	81	0.6467	0.5196	0.0490	
sp_richn	fide_Dim1	5.2479	14.624	81	0.3589	0.7206	0.0302	
sp_richn	feve	-2.7646	17.8268	81	-0.1551	0.8771	-0.0108	
sp_richn	log(dens)	11.7587	1.0676	81	11.0138	0.0000	0.8971	***
log(dens)	hydro	-0.7743	0.2573	82	-3.0095	0.0035	-0.3535	**
log(dens)	mfdis	-6.8959	1.3639	82	-5.0561	0.0000	-0.4993	***
log(dens)	convexity	1.6745	0.9714	82	1.7238	0.0885	0.2119	
log(dens)	sqrt(biomass)	0.0026	0.0033	82	0.7770	0.4394	0.0768	
log(dens)	log(surf)	-1.2618	1.0988	82	-1.1483	0.2542	-0.1170	
log(dens)	fide_Dim1	-3.6822	1.457	82	-2.5273	0.0134	-0.2775	*
log(dens)	feve	0.2942	1.8437	82	0.1596	0.8736	0.0151	
feve	hydro	-0.0454	0.0149	84	-3.0436	0.0031	-0.4036	**
feve	mfdis	-0.0761	0.085	84	-0.8958	0.3729	-0.1072	
feve	convexity	0.1481	0.051	84	2.9067	0.0047	0.3646	**
feve	sqrt(biomass)	-0.0002	2e-04	84	-0.9590	0.3403	-0.1112	
feve	log(surf)	0.1975	0.0653	84	3.0263	0.0033	0.3563	**
fide_Dim1	hydro	-0.0608	0.0189	84	-3.2178	0.0018	-0.3684	**
fide_Dim1	mfdis	-0.1229	0.1075	84	-1.1433	0.2562	-0.1181	
fide_Dim1	convexity	0.3865	0.0645	84	5.9950	0.0000	0.6491	***
fide_Dim1	log(surf)	0.0059	0.0826	84	0.0719	0.9429	0.0073	
fide_Dim1	sqrt(biomass)	0.0005	3e-04	84	2.0733	0.0412	0.2076	*

Response	Predictor	Estimate	Std.Error	DF	Crit.Value	P.Value	Std.Estimate	
mfdis	hydro	0.0707	0.0151	88	4.6756	0.0000	0.4461	***
convexity	hydro	0.1276	0.0262	88	4.8631	0.0000	0.4602	***
sqrt(biomass)	hydro	-19.6145	6.6469	88	-2.9509	0.0041	-0.3001	**
log(surf)	hydro	0.0775	0.0228	87	3.3979	0.0010	0.3816	**
log(surf)	convexity	-0.2334	0.0823	87	-2.8361	0.0057	-0.3185	**
~~log(surf)	~~sqrt(biomass)	0.4016	-	90	4.0906	0.0000	0.4016	***
~~feve	~~fide_Dim1	0.3389	-	90	3.3599	0.0006	0.3389	***
~~mfdis	~~convexity	0.3245	-	90	3.2004	0.0010	0.3245	***

Table D 22: Individual R-squared

Response	family	link	method	R.squared
sp_richn	gaussian	identity	none	0.71
dens	gaussian	identity	none	0.45
feve	gaussian	identity	none	0.16
fide_Dim1	gaussian	identity	none	0.38
mfdis	gaussian	identity	none	0.20
convexity	gaussian	identity	none	0.21
biomass	gaussian	identity	none	0.09
surf	gaussian	identity	none	0.14

pSEM 2

pSEM fit for the model containing only first order orthogonal polynomial terms:

sp_richn ~ hydro + mfdis_1 + convexity + sqrt(biomass) + log(surf) + fide_Dim1 + feve_1 + log(dens),

log(dens) ~ hydro + mfdis_1 + convexity + sqrt(biomass) + log(surf) + fide_Dim1 + feve_1

feve_1 ~ hydro + mfdis_1 + convexity + sqrt(biomass) + log(surf)

fide_Dim1 ~ hydro + mfdis_1 + convexity + log(surf) + sqrt(biomass)

mfdis_1 ~ hydro

convexity ~ hydro

sqrt(biomass) ~ hydro

log(surf) ~ hydro + convexity

log(surf) ~~ sqrt(biomass)

feve_1 ~~ fide_Dim1

mfdis_1 ~~ convexity

Table D 23: Tests of directed separation

Independ.Claim	Test.Type	DF	Crit.Value	P.Value
sqrt(biomass) ~ mfdis_1 + …	coef	87	-1.2835	0.2027
log(surf) ~ mfdis_1 + …	coef	86	0.2374	0.8129
sqrt(biomass) ~ convexity + …	coef	87	-0.2273	0.8207

Table D 24: Global goodness-of-fit

Chisq	df	P.Value	Fisher.C	df	P.Value
1.788	3	0.618	4.001	6	0.677

Table D 25: Model coefficients

Response	Predictor	Estimate	Std.Error	DF	Crit.Value	P.Value	Std.Estimate	
sp_richn	hydro	2.2040	2.6192	81	0.8415	0.4025	0.0768	
sp_richn	mfdis_1	1.3177	8.2463	81	0.1598	0.8734	0.0132	
sp_richn	convexity	-2.9011	9.5229	81	-0.3046	0.7614	-0.0280	
sp_richn	sqrt(biomass)	-0.0159	0.0322	81	-0.4934	0.6231	-0.0361	
sp_richn	log(surf)	6.9536	10.7182	81	0.6488	0.5183	0.0492	
sp_richn	fide_Dim1	4.6761	14.5457	81	0.3215	0.7487	0.0269	
sp_richn	feve_1	-1.0953	6.9743	81	-0.1570	0.8756	-0.0110	
sp_richn	log(dens)	11.6663	1.0449	81	11.1655	0.0000	0.8900	***
log(dens)	hydro	-0.7534	0.264	82	-2.8536	0.0055	-0.3440	**
log(dens)	mfdis_1	-3.5586	0.7779	82	-4.5745	0.0000	-0.4687	***
log(dens)	convexity	1.5334	0.9921	82	1.5456	0.1261	0.1941	
log(dens)	sqrt(biomass)	0.0028	0.0034	82	0.8200	0.4146	0.0829	
log(dens)	log(surf)	-1.2728	1.1241	82	-1.1323	0.2608	-0.1180	
log(dens)	fide_Dim1	-3.5213	1.4874	82	-2.3675	0.0203	-0.2654	*
log(dens)	feve_1	0.0474	0.7371	82	0.0643	0.9489	0.0062	
feve_1	hydro	-0.1137	0.0384	84	-2.9589	0.0040	-0.3941	**
feve_1	mfdis_1	-0.1314	0.1211	84	-1.0853	0.2809	-0.1314	
feve_1	convexity	0.3874	0.1302	84	2.9759	0.0038	0.3724	**
feve_1	sqrt(biomass)	-0.0005	5e-04	84	-0.9851	0.3274	-0.1139	
feve_1	log(surf)	0.5085	0.1668	84	3.0484	0.0031	0.3581	**
fide_Dim1	hydro	-0.0604	0.019	84	-3.1726	0.0021	-0.3660	**
fide_Dim1	mfdis_1	-0.0643	0.06	84	-1.0707	0.2874	-0.1123	
fide_Dim1	convexity	0.3849	0.0645	84	5.9660	0.0000	0.6465	***
fide_Dim1	log(surf)	0.0052	0.0827	84	0.0634	0.9496	0.0064	

Response	Predictor	Estimate	Std.Error	DF	Crit.Value	P.Value	Std.Estimate	
fide_Dim1	sqrt(biomass)	0.0005	3e-04	84	2.0947	0.0392	0.2097	*
mfdis_1	hydro	0.1362	0.0271	88	5.0243	0.0000	0.4721	***
convexity	hydro	0.1276	0.0262	88	4.8631	0.0000	0.4602	***
sqrt(biomass)	hydro	-19.6145	6.6469	88	-2.9509	0.0041	-0.3001	**
log(surf)	hydro	0.0775	0.0228	87	3.3979	0.0010	0.3816	**
log(surf)	convexity	-0.2334	0.0823	87	-2.8361	0.0057	-0.3185	**
~~log(surf)	~~sqrt(biomass)	0.4016	-	90	4.0906	0.0000	0.4016	***
~~feve_1	~~fide_Dim1	0.3377	-	90	3.3465	0.0006	0.3377	***
~~mfdis_1	~~convexity	0.3246	-	90	3.2008	0.0010	0.3246	***

Table D 26: Individual R-squared

Response	family	link	method	R.squared
sp_richn	gaussian	identity	none	0.71
dens	gaussian	identity	none	0.43
feve_1	gaussian	identity	none	0.17
fide_Dim1	gaussian	identity	none	0.37
mfdis_1	gaussian	identity	none	0.22
convexity	gaussian	identity	none	0.21
biomass	gaussian	identity	none	0.09
surf	gaussian	identity	none	0.14

pSEM 3 (final selected model)

pSEM fit for the model containing both first and second order orthogonal polynomial terms:
sp_richn ~ hydro + mfdis_1 + mfdis_2 + convexity + sqrt(biomass) + log(surf) + fide_Dim1 + feve_1 + feve_2 + log(dens)

log(dens) ~ hydro + mfdis_1 + mfdis_2 + convexity + sqrt(biomass) + log(surf) + fide_Dim1 + feve_1 + feve_2

feve_1 ~ hydro + mfdis_1 + mfdis_2 + convexity + sqrt(biomass) + log(surf)

fide_Dim1 ~ hydro + mfdis_1 + mfdis_2 + convexity + log(surf) + sqrt(biomass)

mfdis_1 ~ hydro

convexity ~ hydro

sqrt(biomass) ~ hydro

log(surf) ~ hydro + convexity

log(surf) ~~ sqrt(biomass)

feve_1 ~~ fide_Dim1

mfdis_1 ~~ convexity

Table D 27: Tests of directed separation

Independ.Claim	Test.Type	DF	Crit.Value	P.Value
mfdis_1 ~ mfdis_2 + …	coef	87	0.8218	0.4134
convexity ~ mfdis_2 + …	coef	87	0.5292	0.5980
sqrt(biomass) ~ mfdis_2 + …	coef	87	-0.6517	0.5163
log(surf) ~ mfdis_2 + …	coef	86	-0.0616	0.9510
mfdis_1 ~ feve_2 + …	coef	87	1.4229	0.1583
convexity ~ feve_2 + …	coef	87	-0.1766	0.8602
sqrt(biomass) ~ feve_2 + …	coef	87	1.1930	0.2361
log(surf) ~ feve_2 + …	coef	86	0.6477	0.5189
fide_Dim1 ~ feve_2 + …	coef	82	0.8961	0.3728
feve_1 ~ feve_2 + …	coef	82	0.4301	0.6682
sqrt(biomass) ~ mfdis_1 + …	coef	87	-1.2835	0.2027
log(surf) ~ mfdis_1 + …	coef	86	0.2374	0.8129
sqrt(biomass) ~ convexity + …	coef	87	-0.2273	0.8207

Table D 28: Global goodness-of-fit

Chisq	df	P.Value	Fisher.C	df	P.Value
8.791	13	0.789	19.184	26	0.829

Table D 29: Model coefficients

Response	Predictor	Estimate	Std.Error	DF	Crit.Value	P.Value	Std.Estimate	
sp_richn	hydro	3.0603	2.6864	79	1.1392	0.2581	0.1066	
sp_richn	mfdis_1	2.6640	8.2132	79	0.3244	0.7465	0.0268	
sp_richn	mfdis_2	5.3808	6.4208	79	0.8380	0.4045	0.0541	
sp_richn	convexity	-4.8203	9.5142	79	-0.5066	0.6138	-0.0465	
sp_richn	sqrt(biomass)	-0.0088	0.0322	79	-0.2718	0.7865	-0.0199	
sp_richn	log(surf)	6.9365	10.6576	79	0.6508	0.5170	0.0491	
sp_richn	fide_Dim1	7.3322	14.6657	79	0.5000	0.6185	0.0422	
sp_richn	feve_1	-1.6910	6.9839	79	-0.2421	0.8093	-0.0170	
sp_richn	feve_2	-10.2813	6.471	79	-1.5888	0.1161	-0.1033	
sp_richn	log(dens)	11.4875	1.1248	79	10.2125	0.0000	0.8764	***
log(dens)	hydro	-0.7755	0.2525	80	-3.0706	0.0029	-0.3541	**

Response	Predictor	Estimate	Std.Error	DF	Crit.Value	P.Value	Std.Estimate	
log(dens)	mfdis_1	-3.0618	0.7411	80	-4.1313	0.0001	-0.4032	***
log(dens)	mfdis_2	-1.7103	0.6089	80	-2.8090	0.0062	-0.2252	**
log(dens)	convexity	1.4072	0.9325	80	1.5091	0.1352	0.1781	
log(dens)	sqrt(biomass)	0.0035	0.0032	80	1.1022	0.2737	0.1046	
log(dens)	log(surf)	-1.3562	1.0484	80	-1.2936	0.1995	-0.1258	
log(dens)	fide_Dim1	-3.6357	1.3999	80	-2.5972	0.0112	-0.2740	*
log(dens)	feve_1	0.3290	0.6932	80	0.4746	0.6363	0.0433	
log(dens)	feve_2	-1.5404	0.6197	80	-2.4857	0.0150	-0.2029	*
feve_1	hydro	-0.1072	0.0391	83	-2.7398	0.0075	-0.3715	**
feve_1	mfdis_1	-0.1385	0.1215	83	-1.1400	0.2576	-0.1385	
feve_1	mfdis_2	0.0932	0.1017	83	0.9167	0.3620	0.0932	
feve_1	convexity	0.3833	0.1304	83	2.9391	0.0043	0.3683	**
feve_1	sqrt(biomass)	-0.0005	5e-04	83	-0.9264	0.3569	-0.1074	
feve_1	log(surf)	0.5058	0.167	83	3.0292	0.0033	0.3562	**
fide_Dim1	hydro	-0.0625	0.0194	83	-3.2151	0.0019	-0.3787	**
fide_Dim1	mfdis_1	-0.0620	0.0604	83	-1.0270	0.3074	-0.1083	
fide_Dim1	mfdis_2	-0.0299	0.0505	83	-0.5913	0.5559	-0.0522	
fide_Dim1	convexity	0.3863	0.0648	83	5.9600	0.0000	0.6487	***
fide_Dim1	log(surf)	0.0061	0.083	83	0.0733	0.9417	0.0075	
fide_Dim1	sqrt(biomass)	0.0005	3e-04	83	2.0466	0.0439	0.2060	*
mfdis_1	hydro	0.1362	0.0271	88	5.0243	0.0000	0.4721	***
convexity	hydro	0.1276	0.0262	88	4.8631	0.0000	0.4602	***
sqrt(biomass)	hydro	-19.6145	6.6469	88	-2.9509	0.0041	-0.3001	**
log(surf)	hydro	0.0775	0.0228	87	3.3979	0.0010	0.3816	**
log(surf)	convexity	-0.2334	0.0823	87	-2.8361	0.0057	-0.3185	**
~~log(surf)	~~sqrt(biomass)	0.4016	-	90	4.0906	0.0000	0.4016	***
~~feve_1	~~fide_Dim1	0.3467	-	90	3.4471	0.0004	0.3467	***
~~mfdis_1	~~convexity	0.3246	-	90	3.2008	0.0010	0.3246	***

Table D 30: Individual R-squared

Response	family	link	method	R.squared
sp_richn	gaussian	identity	none	0.72
dens	gaussian	identity	none	0.52
feve_1	gaussian	identity	none	0.17
fide_Dim1	gaussian	identity	none	0.38
mfdis_1	gaussian	identity	none	0.22
convexity	gaussian	identity	none	0.21
biomass	gaussian	identity	none	0.09
surf	gaussian	identity	none	0.14

pSEM 4

pSEM fit for the model containing the generalised linear path model for nodule convexity and both first and second order orthogonal polynomial terms:
sp_richn ~ hydro + mfdis_1 + mfdis_2 + convexity + sqrt(biomass) + log(surf) + fide_Dim1 + feve_1 + feve_2 + log(dens)

log(dens) ~ hydro + mfdis_1 + mfdis_2 + convexity + sqrt(biomass) + log(surf) + fide_Dim1 + feve_1 + feve_2

feve_1 ~ hydro + mfdis_1 + mfdis_2 + convexity + sqrt(biomass) + log(surf)

fide_Dim1 ~ hydro + mfdis_1 + mfdis_2 + convexity + log(surf) + sqrt(biomass)

mfdis_1 ~ hydro

convexity ~ hydro family = Gamma(link = "inverse")

sqrt(biomass) ~ hydro

log(surf) ~ hydro + convexity

log(surf) ~~ sqrt(biomass)

feve_1 ~~ fide_Dim1

mfdis_1 ~~ convexity

Table D 31: Global goodness-of-fit

Chisq	df	P.Value	Fisher.C	df	P.Value
8.762	13	0.791	19.126	26	0.831

Table D 32: Model coefficients

Response	Predictor	Estimate	Std.Error	DF	Crit.Value	P.Value
sp_richn	hydro	3.0602713	2.6864355	79	1.1391568	0.2580802
sp_richn	mfdis_1	2.6639872	8.2132204	79	0.3243536	0.7465276
sp_richn	mfdis_2	5.3807945	6.4207717	79	0.8380293	0.4045426
sp_richn	convexity	-4.8202619	9.5141548	79	-0.5066411	0.6138178
sp_richn	sqrt(biomass)	-0.0087620	0.0322330	79	-0.2718335	0.7864592
sp_richn	log(surf)	6.9364538	10.6575861	79	0.6508466	0.5170341
sp_richn	fide_Dim1	7.3321655	14.6656921	79	0.4999536	0.6184975
sp_richn	feve_1	-1.6910095	6.9838820	79	-0.2421303	0.8093067
sp_richn	feve_2	-10.2812773	6.4710436	79	-1.5888129	0.1160971

Response	Predictor	Estimate	Std.Error	DF	Crit.Value	P.Value	
sp_richn	log(dens)	11.4874631	1.1248470	79	10.2124675	0.0000000	***
log(dens)	hydro	-0.7754673	0.2525488	80	-3.0705639	0.0029183	**
log(dens)	mfdis_1	-3.0617537	0.7411100	80	-4.1313083	0.0000881	***
log(dens)	mfdis_2	-1.7102906	0.6088683	80	-2.8089666	0.0062440	**
log(dens)	convexity	1.4072159	0.9324732	80	1.5091221	0.1352071	
log(dens)	sqrt(biomass)	0.0035046	0.0031797	80	1.1021735	0.2736911	
log(dens)	log(surf)	-1.3561988	1.0483954	80	-1.2935947	0.1995280	
log(dens)	fide_Dim1	-3.6357134	1.3998642	80	-2.5971901	0.0111831	*
log(dens)	feve_1	0.3290116	0.6931829	80	0.4746389	0.6363375	
log(dens)	feve_2	-1.5403695	0.6196997	80	-2.4856709	0.0150148	*
feve_1	hydro	-0.1071633	0.0391139	83	-2.7397745	0.0075246	**
feve_1	mfdis_1	-0.1384547	0.1214506	83	-1.1400089	0.2575617	
feve_1	mfdis_2	0.0932249	0.1016953	83	0.9167088	0.3619512	
feve_1	convexity	0.3832610	0.1303989	83	2.9391442	0.0042612	**
feve_1	sqrt(biomass)	-0.0004738	0.0005115	83	-0.9263530	0.3569484	
feve_1	log(surf)	0.5058313	0.1669829	83	3.0292407	0.0032674	**
fide_Dim1	hydro	-0.0625053	0.0194411	83	-3.2151158	0.0018585	**
fide_Dim1	mfdis_1	-0.0619976	0.0603654	83	-1.0270374	0.3073867	
fide_Dim1	mfdis_2	-0.0298875	0.0505463	83	-0.5912902	0.5559330	
fide_Dim1	convexity	0.3862840	0.0648131	83	5.9599711	0.0000001	***
fide_Dim1	log(surf)	0.0060848	0.0829967	83	0.0733136	0.9417330	
fide_Dim1	sqrt(biomass)	0.0005203	0.0002542	83	2.0465632	0.0438656	*
mfdis_1	hydro	0.1361866	0.0271054	88	5.0243332	0.0000026	***
convexity	hydro	-1.0007440	0.2003114	88	-4.9959421	0.0000029	***
sqrt(biomass)	hydro	-19.6144777	6.6468869	88	-2.9509269	0.0040597	**
log(surf)	hydro	0.0775106	0.0228116	87	3.3978544	0.0010263	**
log(surf)	convexity	-0.2333818	0.0822903	87	-2.8360781	0.0056797	**
~~log(surf)	~~sqrt(biomass)	0.4016331	NA	90	4.0906114	0.0000479	***

Response	Predictor	Estimate	Std.Error	DF	Crit.Value	P.Value	
~~feve_1	~~fide_Dim1	0.3466542	NA	90	3.4471208	0.0004375	***
~~mfdis_1	~~convexity	0.3162181	NA	90	3.1090211	0.0012685	**

Direct, Indirect and Net effects

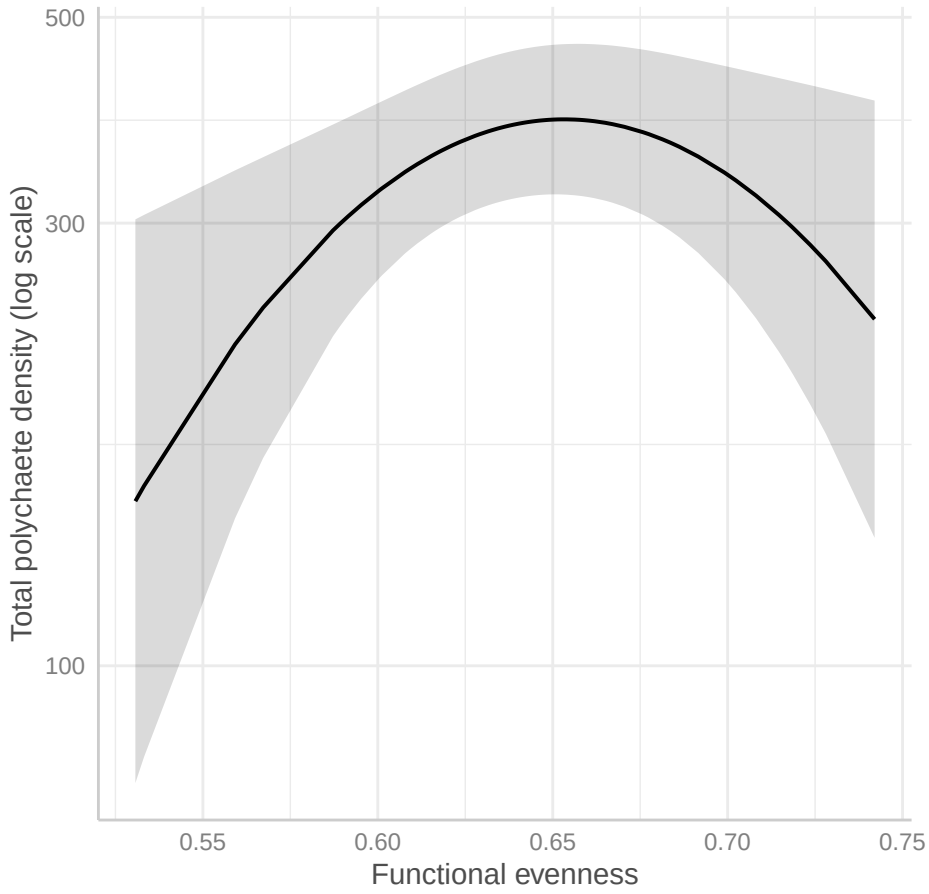
Besides the direct effects reported, indirect and net effects can be calculated in piecewise SEM. Indirect effects can be quantified by the product of standard coefficients of each significant intermediate path in a pathway between a predictor and a response variable. If there are multiple pathways from a predictor and a response variable, the indirect is the sum of the products of each pathway.

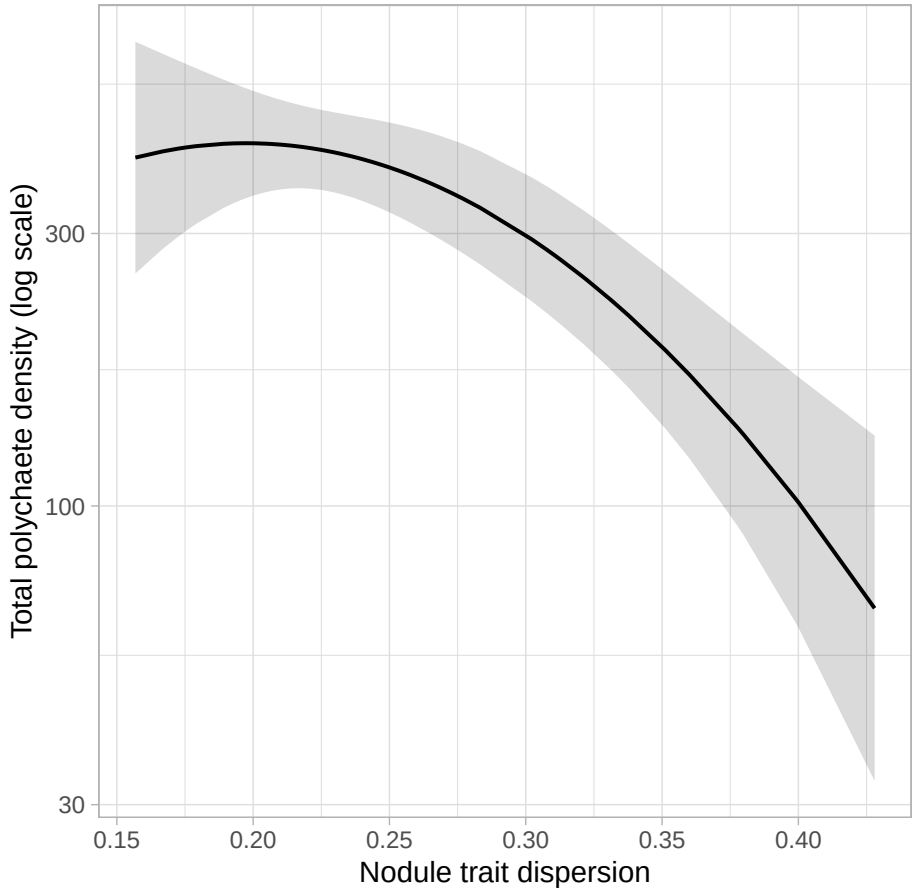
In order to estimate the overall magnitude of the non-linear effects of nodule trait dispersion and functional evenness on total density, the sum of the absolute value of the standard coefficients of both 1st and 2nd order polynomial terms was taken (***0.63*** for nodule trait dispersion, and ***0.25*** for functional evenness). Because those effects are neither positive nor negative, intermediate and net effects are not reported for pathways that include paths upstream nodule trait dispersion or functional evenness (e.g. those leading to total density and species richness).

Table D 33: Direct, Indirect and Net effects

Response	Predictor	Direct	Indirect	Net
Nodule surface area	Hydrosedimentary stress	0.38	-0.15	0.24
Functional identity	Hydrosedimentary stress	-0.38	0.24	-0.14
Functional evenness	Hydrosedimentary stress	-0.37	0.31	-0.07
Functional evenness	Nodule convexity	0.37	-0.11	0.25

Shape of significant non-linear relationships





References

Chai, T., and R. R. Draxler. 2014. "Root Mean Square Error (RMSE) or Mean Absolute Error (MAE)? – Arguments Against Avoiding RMSE in the Literature." *Geoscientific Model Development* 7 (3): 1247–50. https://doi.org/10.5194/gmd-7-1247-2014.

Lefcheck, Jonathan S. 2016. "piecewiseSEM: Piecewise Structural Equation Modelling in r for Ecology, Evolution, and Systematics." *Methods in Ecology and Evolution* 7 (5): 573–79. https://doi.org/10.1111/2041-210X.12512.

Shipley, Bill. 2013. "The AIC Model Selection Method Applied to Path Analytic Models Compared Using a d-Separation Test." *Ecology* 94 (3): 560–64. https://doi.org/10.1890/12-0976.1.

Shipley, Bill, and Jacob C. Douma. 2020. "Generalized AIC and Chi-Squared Statistics for Path Models Consistent with Directed Acyclic Graphs." *Ecology* 101 (3): e02960. https://doi.org/10.1002/ecy.2960.